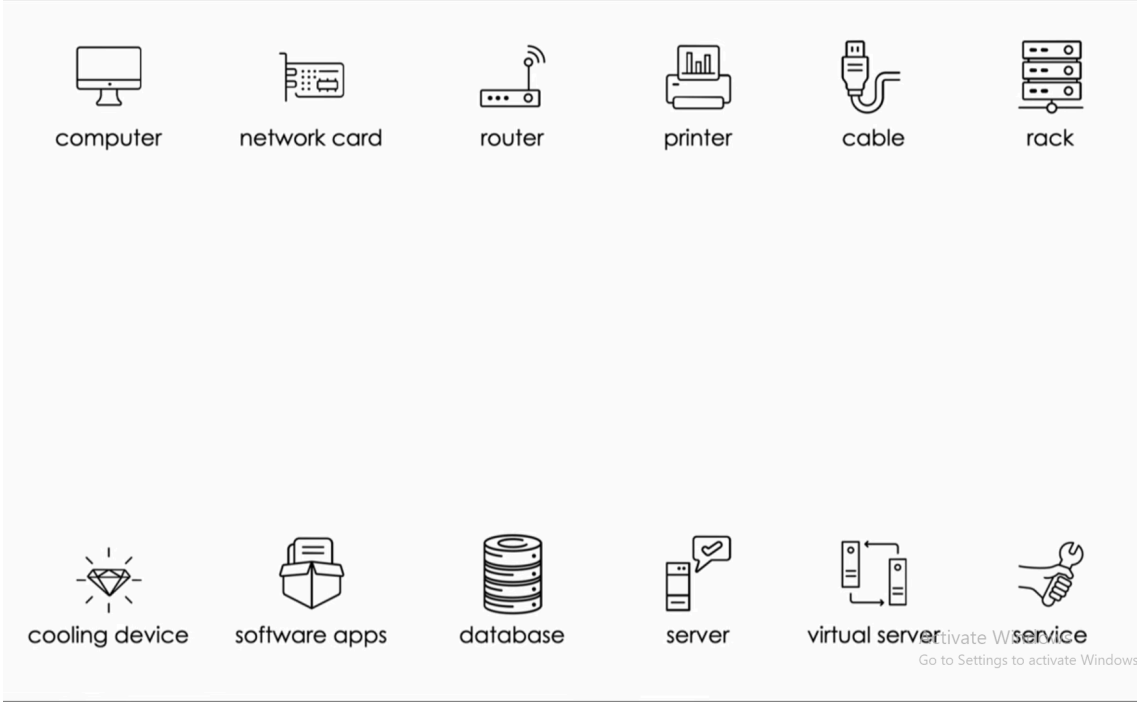
Configuration management is a process that enables company to understand from an operations perspective that how technical components are configured and connected with each other to provide value in the form of IT Services.

A **CMDB** is a set of database tables that store a logical model of company's IT components, the services they provided and the relationships between them.



Each record in the cmdb represents one of these IT components and gets stored in a table that's designed to store the attributes of that item. Each record in cmdb is called CI.

Four Important tables :

1. Base configuration item (cmdb)
2. Configuration item (cmdb_ci)
3. CI Relationship (cmdb_rel_ci)
4. CI Relationship type (cmdb_rel_type)

Populating the CMDB :

Manual Entry

- Manually populate and submit forms
- A form exists for every type of CI
- Time-consuming and error-prone

Auto Discovery

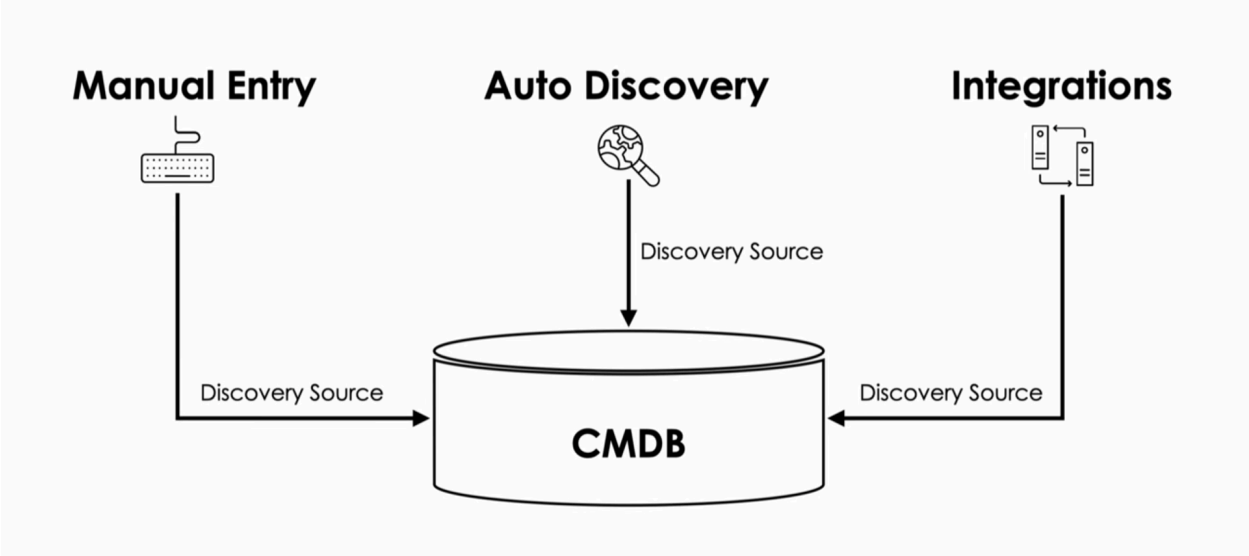
- Network scanning tools identify and load CIs

- SN Discovery, SN Service Mapping, Microsoft SCCM, etc.
- Method of choice for CMDB population

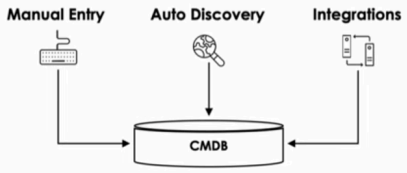
Integrations

- Import from legacy inventory systems
- Data Source, Transform Map, Field Maps, Scheduled Imports
- Great for initial load

Each source of data entering the CMDB is called discovery source. And in most of the cases a single cmdb will have multiple discovery source configured.



Anytime you have multiple sources loading the same database, you run the risk of conflict and inaccuracy.

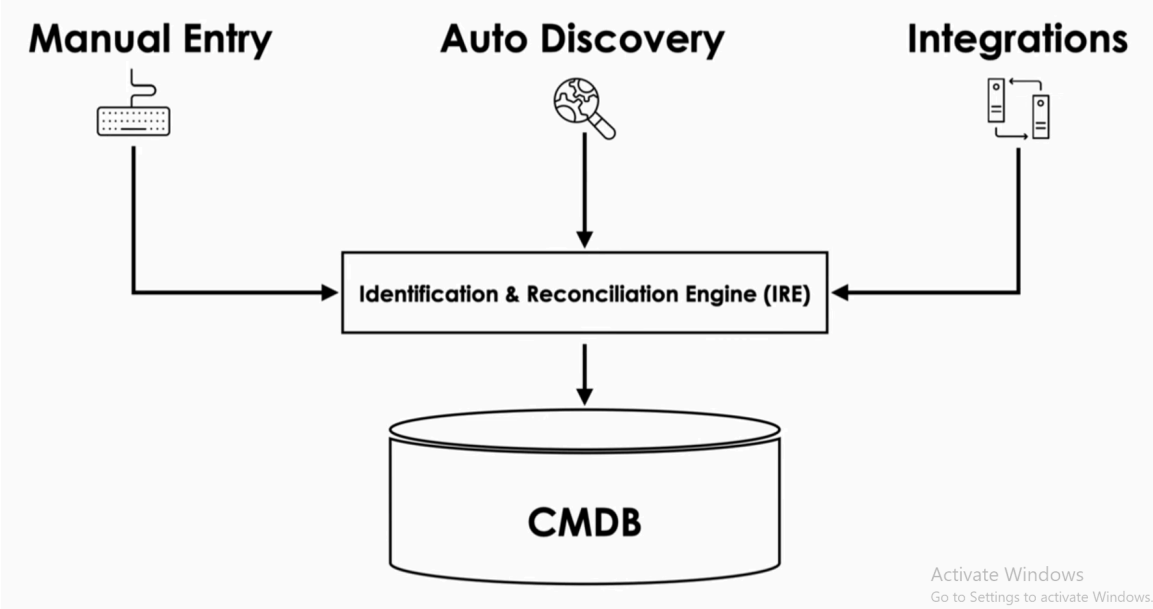


Duplication occurs when a single real-world item is represented by multiple CIs.

Overloading occurs when multiple real-world items are represented by a single CI.

Flip-Flopping occurs when a CI's attributes are updated with different values from different Discovery Sources.

To solve all these problems, all data entering the cmdb must first pass through, what snow calls, the identification and reconciliation engine (IRE).

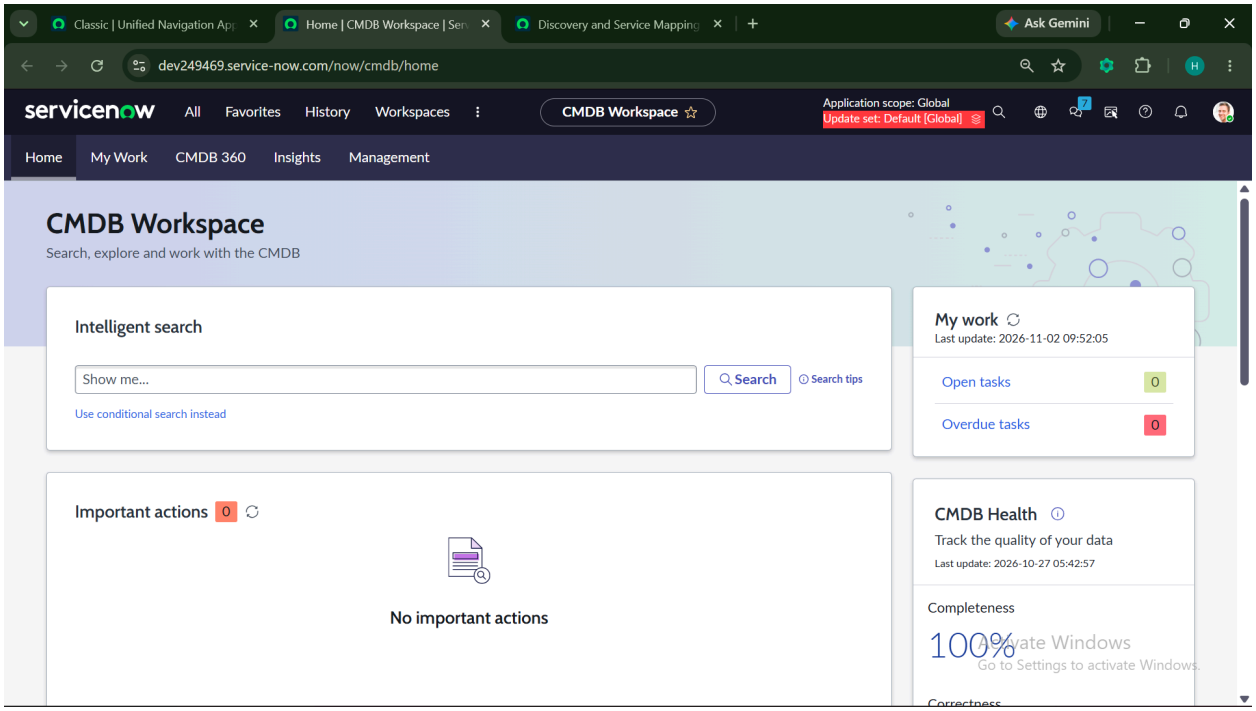


The **Identification & Reconciliation Engine (IRE)** is a set of configurable rules that uniquely identify items and manage updates in the CMDB to avoid duplication, overloading and flip-flopping.

Key Functions of IRE:

1. **Identification Rules:**
    - Uses prioritized criteria (such as MAC address, serial number, IP address, or hostname) to evaluate incoming data.
    - Checks whether the incoming data represents an existing CI or a new one to prevent **duplicate records**.
2. **Reconciliation Rules:**
    - Determines which data source is authorized to update specific attributes of a CI when multiple sources report data for the same item.
    - Prevents less authoritative data sources from overwriting accurate, high-priority data (**flip-flopping** or **overloading**).
3. **Data Integrity & Consistency:**
    - Ensures that all incoming data follows unified validation rules across manual, automated, or integrated data pipelines, maintaining a single source of truth within the CMDB.

CMDB WORKSPACE



What is CMDB Workspace?

CMDB Workspace is a modern ServiceNow screen where you can **view, search, fix, and manage your CMDB** in one place. The CMDB is the database that stores all your IT items (servers, PCs, applications, services), called **CIs**, and how they are connected.

Instead of opening many separate pages, you get one workspace for all CMDB work.

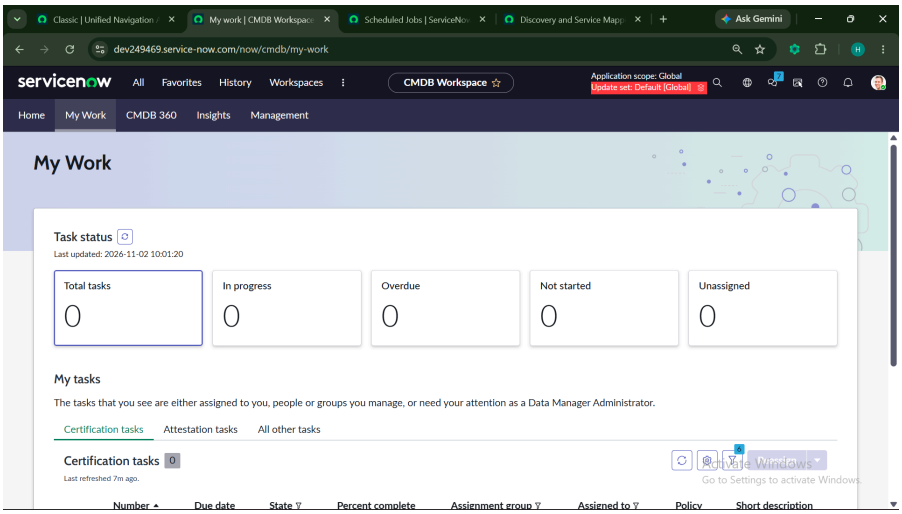
What can we do here?

Area	What you can do
Home	Search CIs in plain language, check CMDB health, see new CIs, and use quick links
My Work	See and finish tasks assigned to you (fix data, duplicates, stale CIs)

CMDB 360	See full details of a CI and its relationships
Insights	Check how well CMDB features are used and get tips to improve
Management	Manage classes, groups, de-duplication, remediation rules, and create services

Main things you can do

1. **Search** any CI quickly
2. **Check data quality** (Completeness, Correctness, Compliance)
3. **Find and fix** duplicate, stale, and orphan CIs
4. **See relationships** between CIs using Dependency Views and the Unified Map
5. **Manage CI classes and groups**
6. **Create** new CIs and services (Business Application, Technical Service)
7. **Track activity** of new and updated CIs



My Work shows CMDB clean-up tasks assigned to you, such as:

- **Health tasks**: fix missing, wrong, or rule-breaking CI data
- **Duplicate CI tasks**: merge or remove copies of the same CI
- **Stale CI tasks**: retire or archive old, unused CIs
- **Orphan CI tasks**: fix CIs with no relationships
- **Certification tasks**: confirm your CI data is still correct

1. Certification Tasks

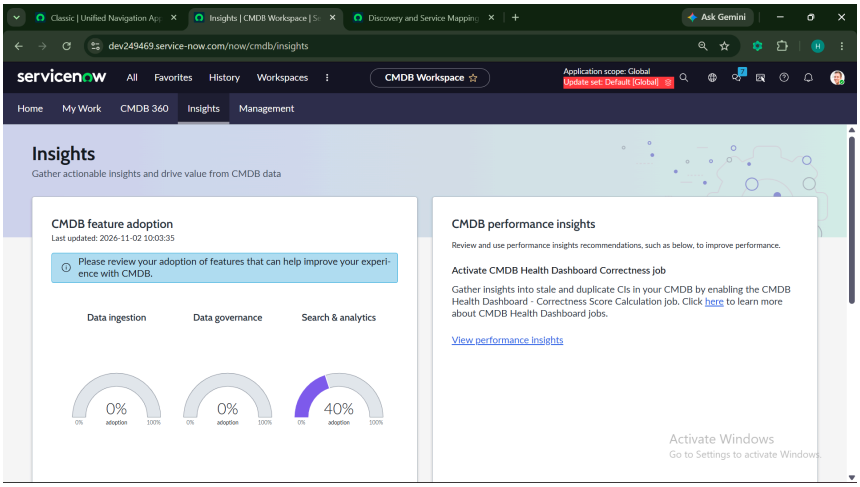
A task asking you to check **specific field values** of a record and mark them as correct or wrong.

- **Example:** System sends you a task: *"Check Server-A. Is the RAM still 32GB, and is the IP 192.168.1.1?"* You inspect it, check "Yes", and complete the task.

2. Attestation Tasks

A task asking you (usually a CI Owner) to confirm that **an entire asset still exists and is in active use**.

- **Example:** System asks: *"Do you still own and use Laptop-102?"* You click **Attest (Yes)** to confirm it hasn't been lost or discarded.



Insights shows how well your CMDB is being used and how to improve it. In ServiceNow CMDB, **Insights** is a dashboard that acts as a **health and usage report card** for your CMDB data. It tells you two simple things: how well you are using ServiceNow's features and what needs fixing to keep your database clean.

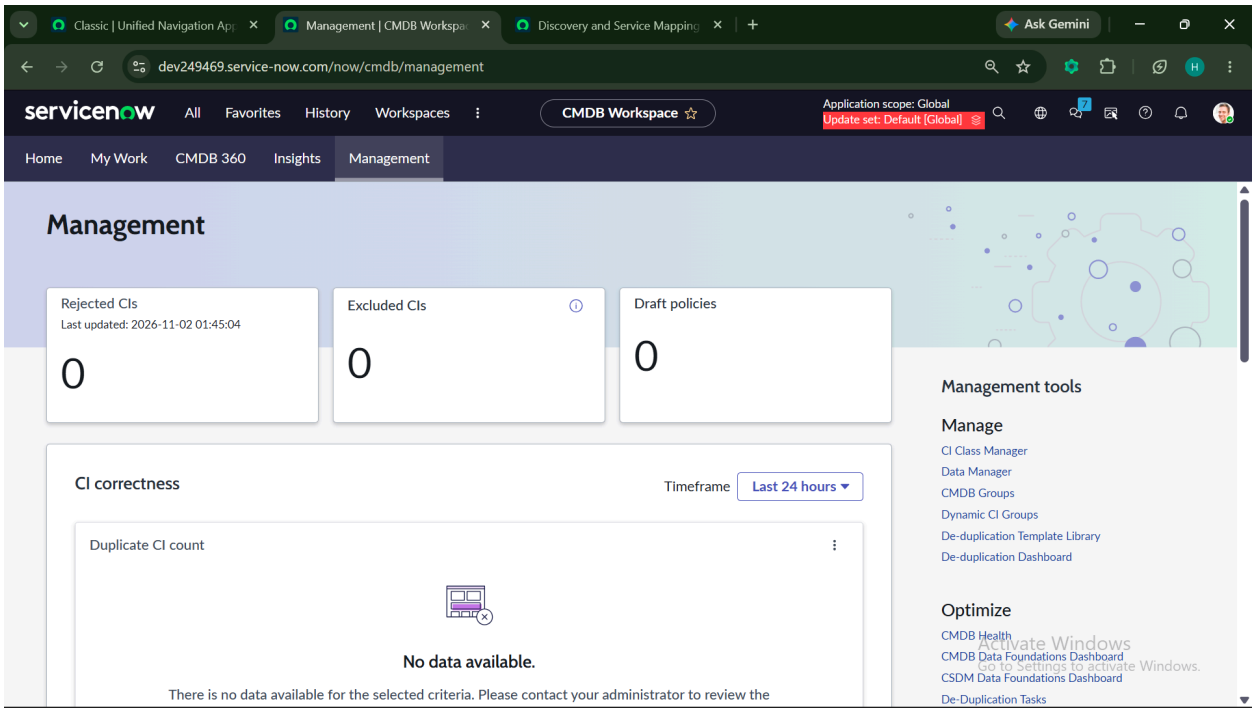
CMDB feature adoption: Tracks how much of ServiceNow's full capabilities you are actually using across three areas:

- **Data ingestion:** How you import data (e.g., Discovery tools).
- **Data governance:** How you manage, clean, and verify data rules.
- **Search & analytics:** How often you query and analyze your data (currently at **40%** in your screenshot).

CMDB performance insights: Gives actionable recommendations to clean up bad or old data. It tells you what is wrong with your database and gives you direct links to fix it.

What it's used for: It stops your system from becoming messy, slow, or outdated.

Real Example: If you have 10 old servers that were turned off months ago, it flags them as "stale" and gives you a 1-click option to remove or archive them.



Management is where admins manage CMDB data and tools.

Top cards

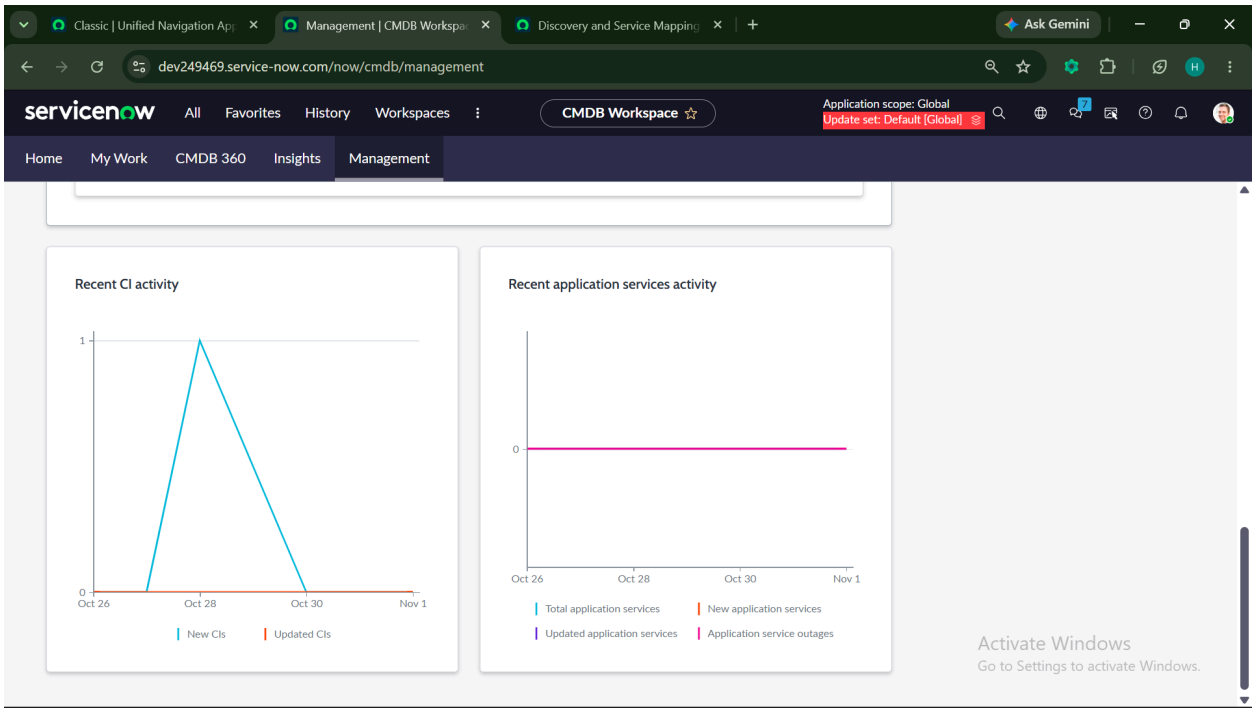
- **Rejected CIs**: CIs that failed checks and were not added (0).
- **Excluded CIs**: CIs left out of health scoring on purpose (0).
- **Draft policies**: health or data rules saved but not yet active (0).

CI correctness

- **Duplicate CI count**: how many copies of the same CI exist. "No data" because the Correctness job is off.
- **Orphan CI count**: CIs with no relationships (0).
- **Timeframe**: change the time period shown.

Management tools (right side)

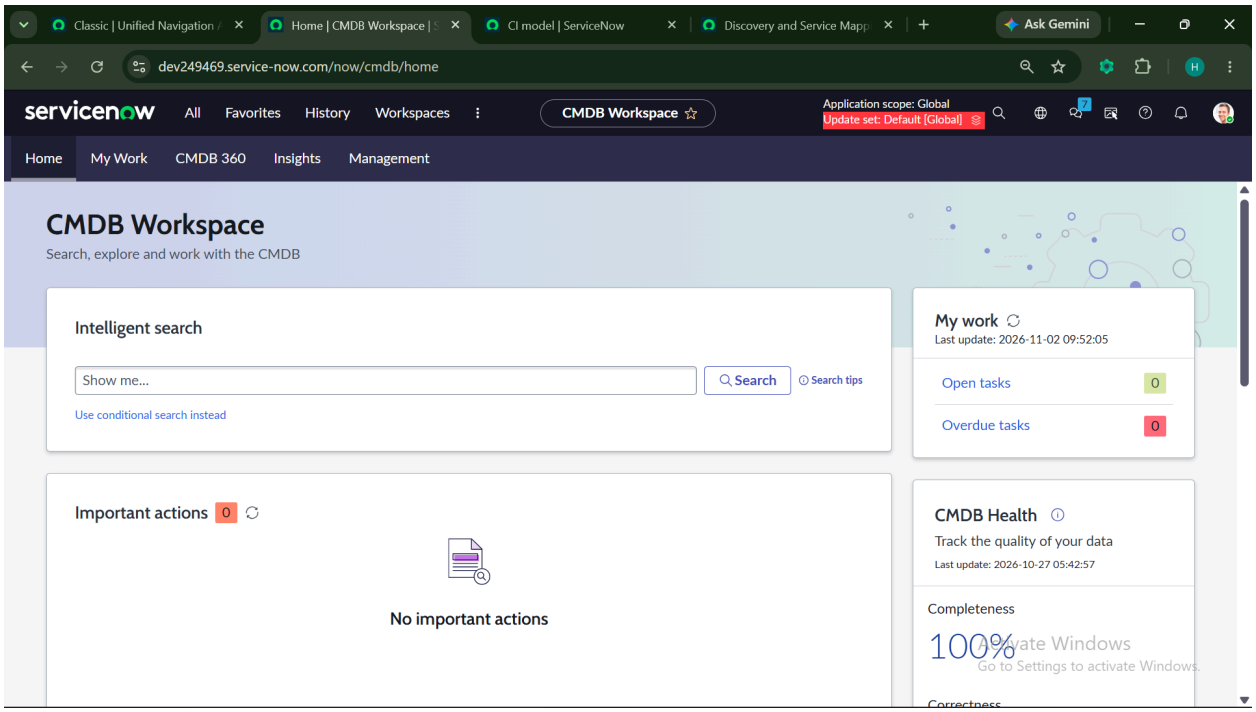
- **Manage**
  - *CI Class Manager*: view and edit CI classes.
  - *Data Manager*: retire or archive old CIs.
  - *CMDB Groups / Dynamic CI Groups*: group CIs by rules.
  - *De-duplication Template Library / Dashboard*: set up and track duplicate removal.
- **Optimize**
  - *CMDB Health*: data quality scores.
  - *Data Foundations Dashboards*: check data setup quality.
  - *De-Duplication Tasks*: fix duplicates.
  - *CMDB Remediation Rules*: auto-create tasks for bad data.
- **Visualize**
  - *Dependency View*, *Query Builder*, *Unified Map*: see and search CI connections.
- **Create**
  - *Application Service*, *Business Application*, *Technical Service*: add new services.

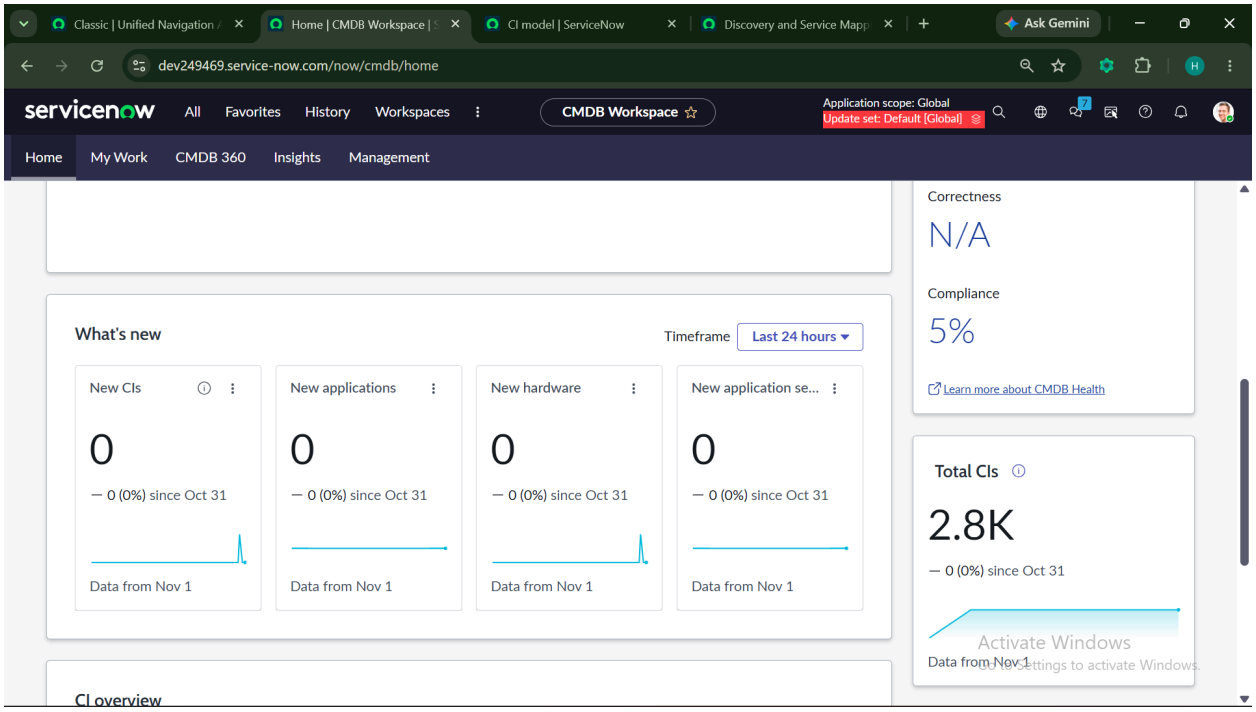


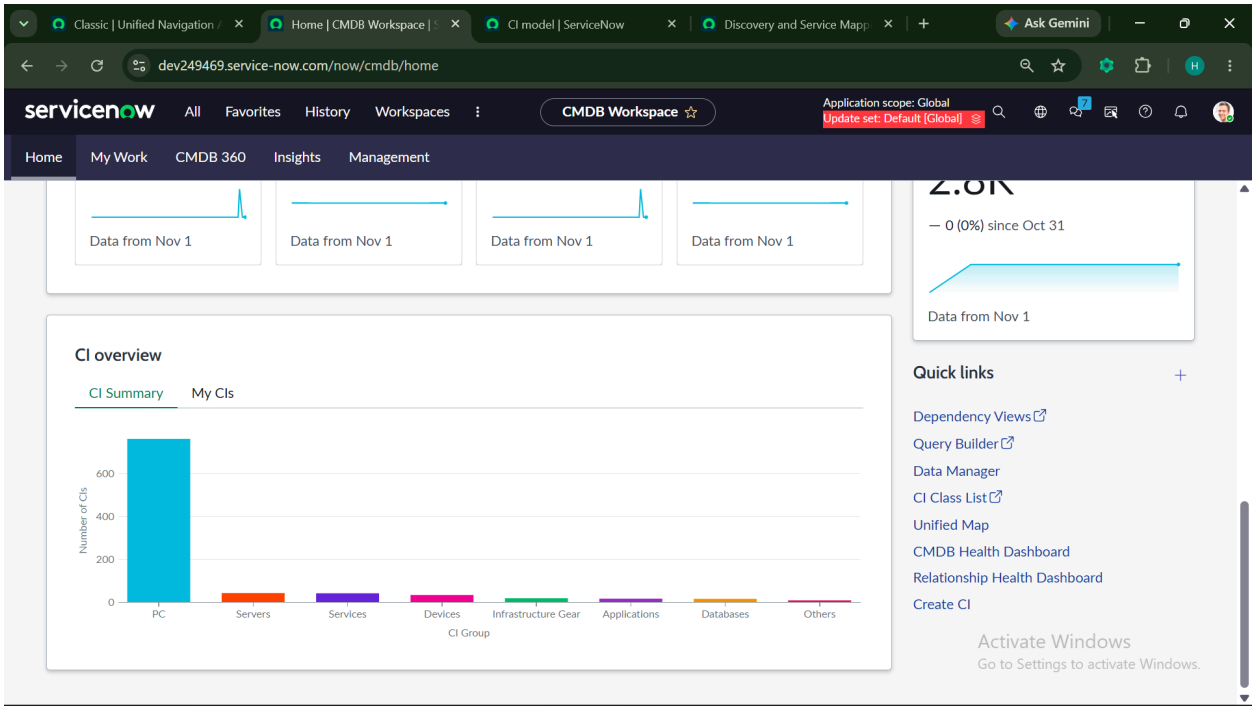
These are two activity charts at the bottom of **Management**:

- **Recent CI activity**: shows new CIs (blue) and updated CIs (orange) over the last few days. Here, 1 new CI was added around Oct 27-28.
- **Recent application services activity**: shows application services over time:
  - *Total*: all services
  - *New*: newly added
  - *Updated*: changed recently
  - *Outages*: services that had downtime

- All are 0 here, so nothing happened.





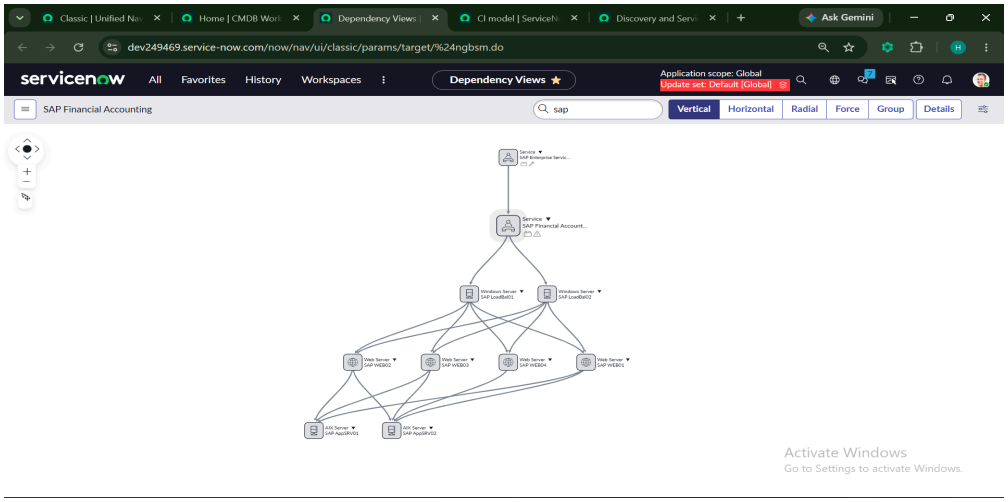


From **Home** you can:

- **Search CIs**: type a plain sentence in Intelligent search
- **Check your tasks**: open or overdue work in My work
- **Check data quality**: Completeness, Correctness, Compliance scores
- **See what's new**: new CIs, applications, hardware, services (change the timeframe)
- **See total CIs**: currently 2.8K
- **View CI breakdown**: chart by group (PC, Servers, Services, etc.)
- **Open tools fast** via Quick links: Dependency Views, Query Builder, Data Manager, CI Class List, Unified Map, Health Dashboards, Create CI

Analysis of its quick links :

1. **Dependency Views :** It shows how a CI or service connects to other CIs, as a map.



In your screen, SAP Financial Accounting (service) depends on:

- 2 Windows load balancers
- 4 web servers
- 2 AIX app servers

What it helps with:

- See what supports a service
- Find what breaks if a CI fails (impact)
- Find the root cause of an issue

Buttons on top:

- **Vertical / Horizontal / Radial / Force / Group**: change the map layout
- **Details**: see info about the selected CI
- **Search box**: find another CI or service

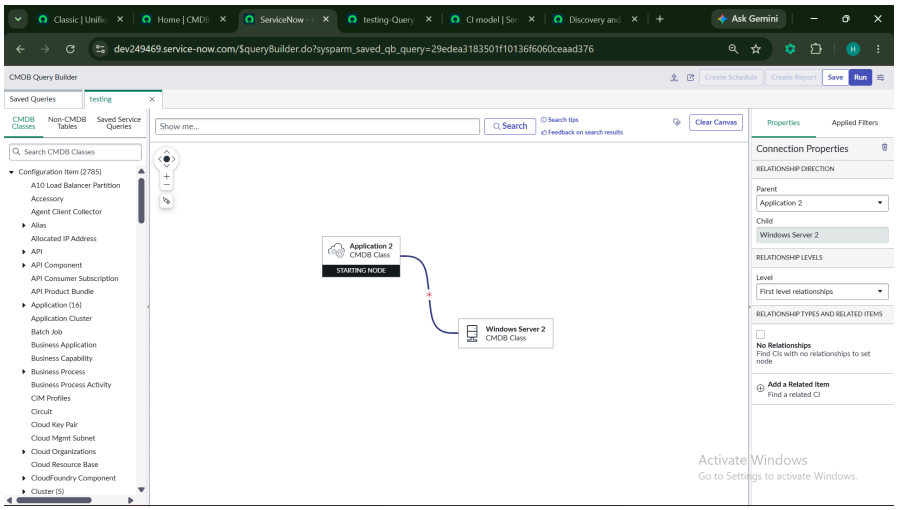
2. **CMDB Query Builder:** lets you build custom searches on CIs and their relationships, using a drag-and-drop screen (no code). You can save the query and rerun it anytime.

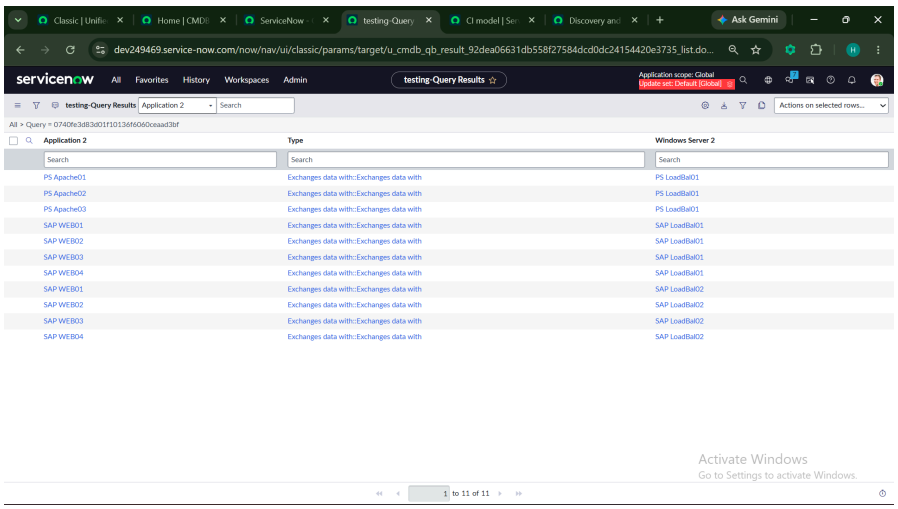
   Uses:

   - Find CIs that match your conditions
   - Find CIs through their relationships (e.g. servers that run an application)
   - Save and reuse queries
   - Export or view results as a list

   Example query you can try (easy):

Show all Windows Servers that run an application





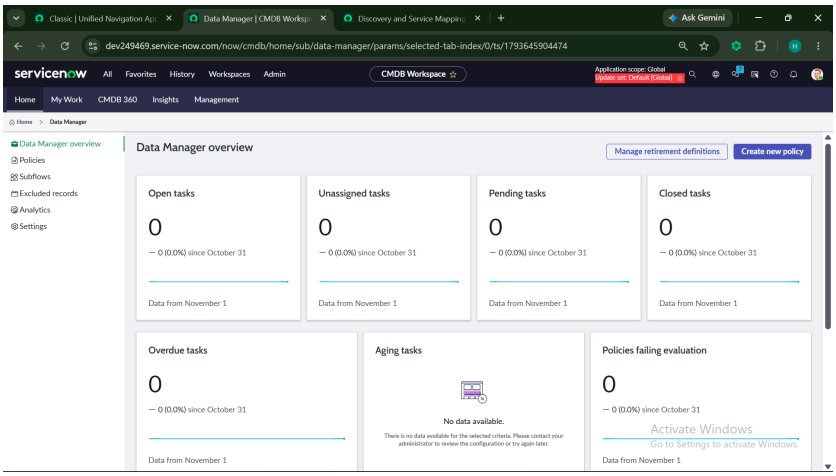
3. **Data Manager:** It is a tool in CMDB Workspace that helps you manage the end of life of CIs. You create policies (rules) that find old, unused, or stale CIs and then retire, archive, or delete them, usually after a task is created for the owner to review.

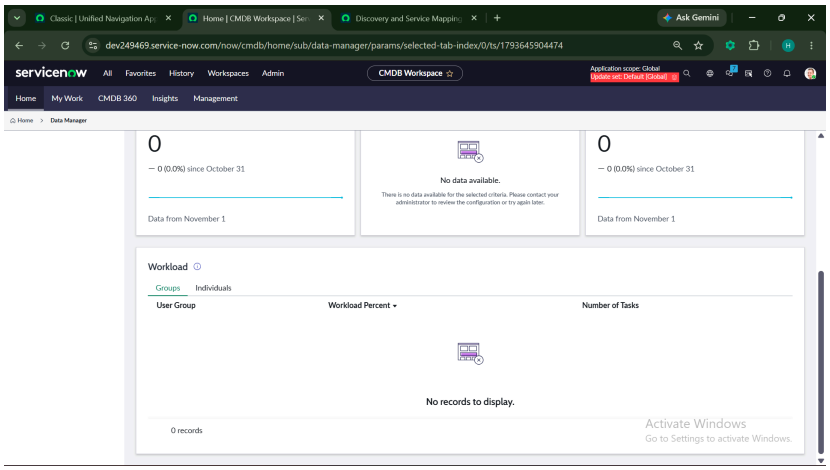
   Purpose: keep the CMDB clean, accurate, and up to date.

   Key parts:

   - Policies: rules that decide which CIs to clean up
   - Retirement definitions: what “retire” means for each CI class
   - Subflows: automatic steps a policy runs
   - Tasks: review work created for owners
   - Excluded records: CIs the policy must skip

- Analytics: reports on results





Let's use one example: **old servers that nobody uses anymore.**

The story

Your CMDB has 2.8K CIs. Some servers were shut down long ago, but their records are still there. These are **stale CIs**. They make your data dirty.

Step by step

1. Policy (the rule)
You click **Create new policy** and write a rule:
"If a Windows Server was not updated in 90 days, retire it."

2. Retirement definition (what "retire" means)
In **Manage retirement definitions**, you set that for servers, "retire" means: change *Operational status* to **Retired**.

3. Subflow (the automatic steps)
The policy runs steps by itself:

- find old servers
- create a task for the owner
- wait for approval
- retire the CI

4. Tasks (the work created)
Say the policy finds 10 old servers. The cards update:

- **Open tasks: 10** (just created)
- **Unassigned tasks: 2** (no owner found)
- **Pending tasks: 3** (waiting for approval)
- **Closed tasks: 5** (done)
- **Overdue tasks: 1** (owner didn't act in time)
- **Aging tasks**: shows that 1 task has been waiting for 30 days

5. Excluded records
One of those servers is a backup server that looks unused but is needed. You add it here, so the policy skips it.

6. Policies failing evaluation
If your rule has a mistake (wrong field name), the policy fails. This card shows **1**, so you can fix it.

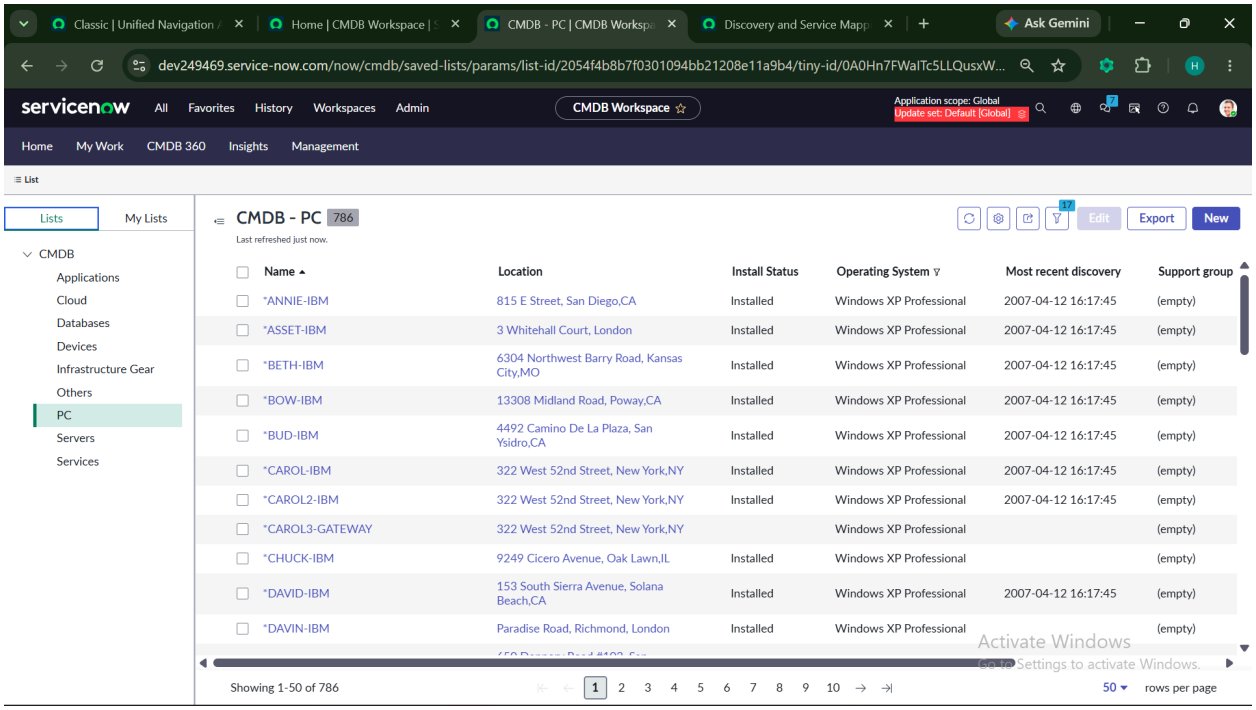
7. Workload
Shows who is busy. Example: *Server Team: 6 tasks (60%)*, *Network Team: 4 tasks (40%)*.

8. Analytics
After a month, you see a report: *"50 CIs retired, 5 excluded"*.

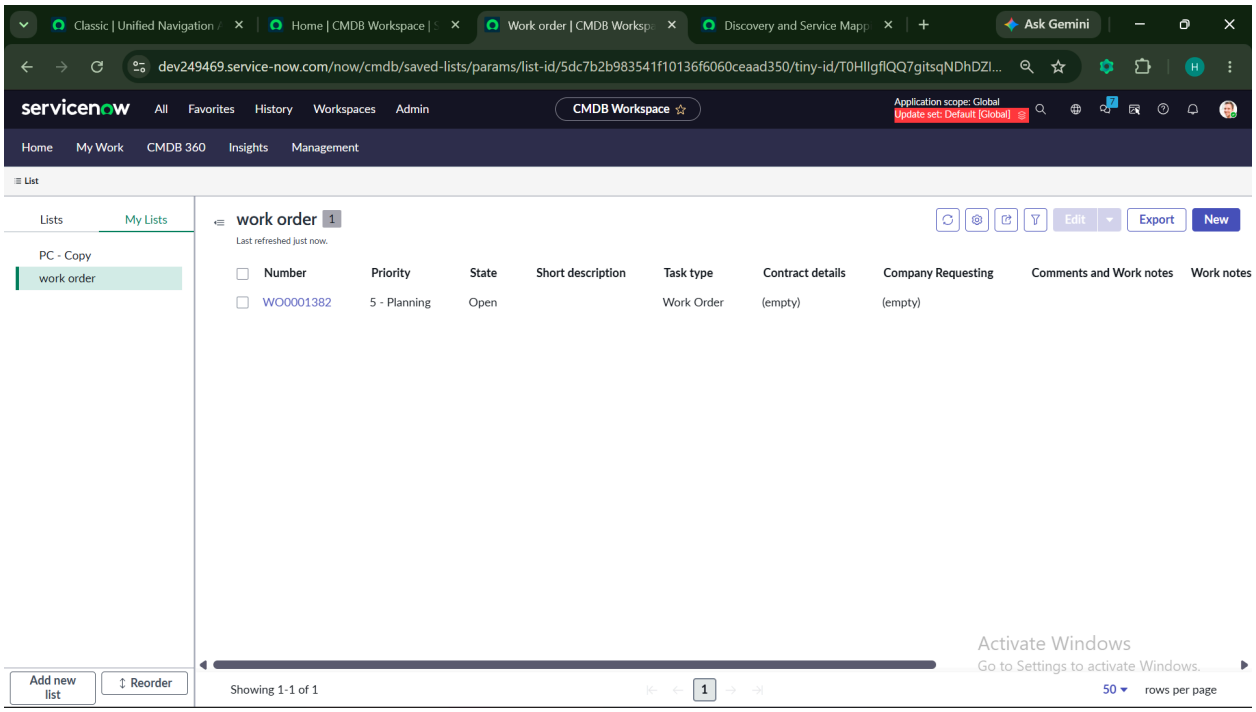
Result

Old records are cleaned up, so your **CMDB Health score** improves.

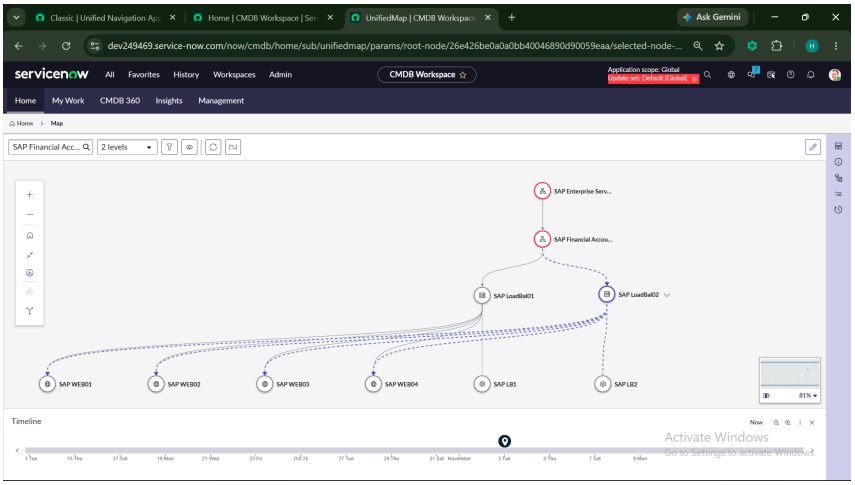
4. **CI Class List:** It is a set of saved lists that shows all CIs grouped by type (Applications, Cloud, Databases, Devices, Servers, PC, and so on). Click a group on the left to see its CIs on the right.



We can also create custom lists of our custom tables:



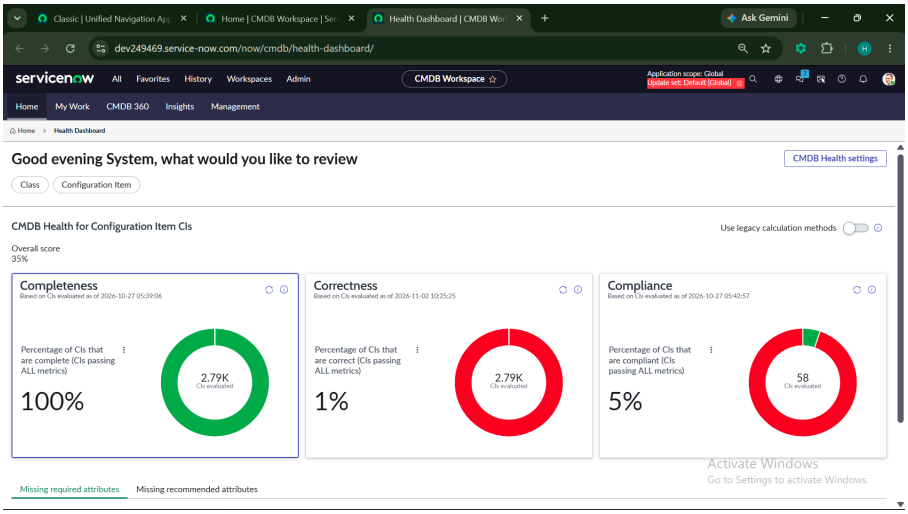
5. **Unified Map:** It is a visual map that shows a CI and everything connected to it (servers, apps, services, and their relationships) in one view. It is like Dependency Views, but newer and inside CMDB Workspace.

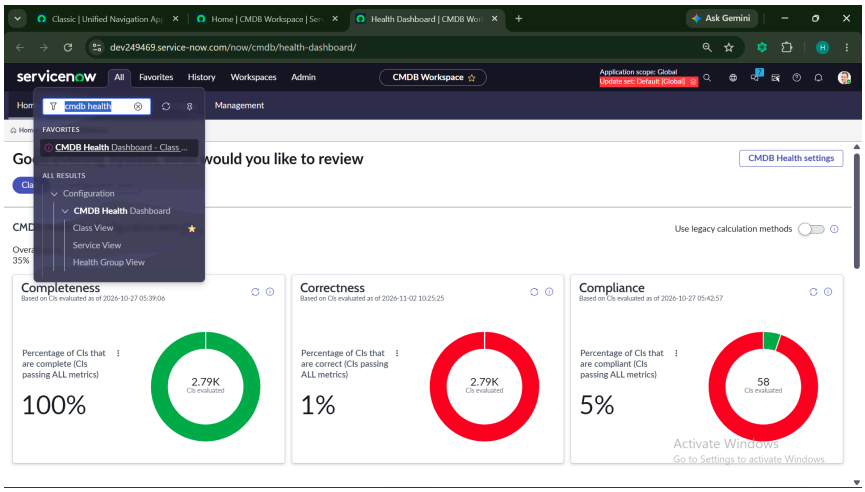


6. **CMDB Health Dashboard**: It shows how good your CMDB data is(basically checks quality of cmdb data), using 3 scores and an overall score (mine: **35%**).

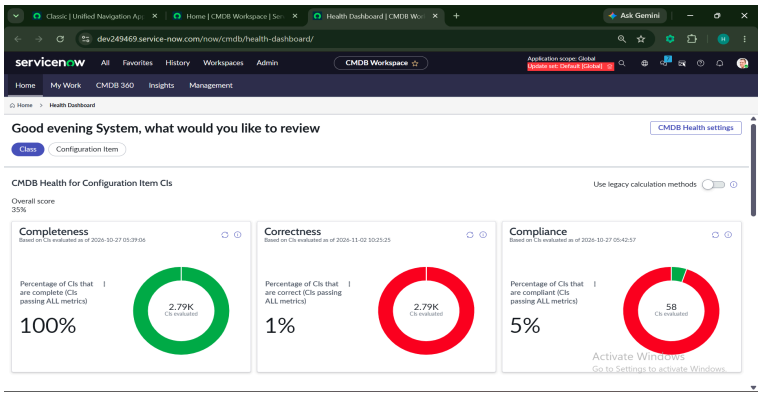
Example: one Windows Server CI

Score	Question it asks	Example	Your result
Completeness	Are important fields filled?	Server has Name, OS, Location, Owner filled = complete. Owner empty = incomplete.	**100%**
Correctness	Is the data unique and not old?	Same server saved twice (duplicate), or not updated for months (stale), or no relationships (orphan) = incorrect.	**1%**
Compliance	Does data follow your rules?	Rule: "OS must be Windows Server 2019 or newer." A server with Windows XP breaks the rule.	**5%**

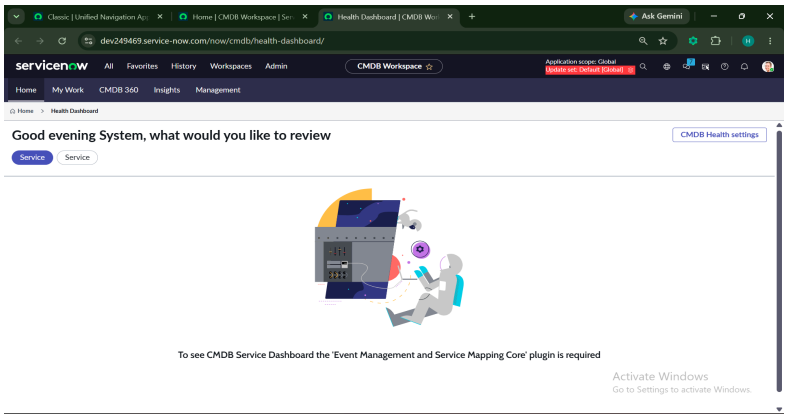




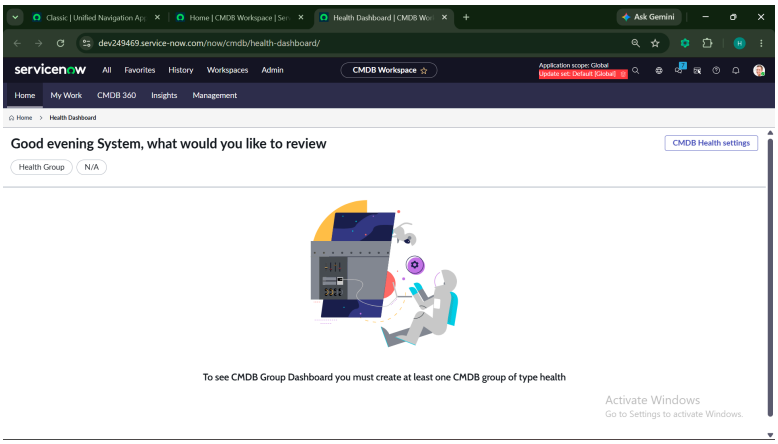
Class view : scores grouped by CI class (Server, PC, Application). Shows which class has the worst data.



Service view : scores for the CIs that support a business service (like SAP Financial Accounting). Shows if a service is at risk.



Health group view : scores by owner group. Shows which team has the most data to fix.



Question: where can i set the rules for correctness, completeness and compliance according to my company's cmdb data policies?

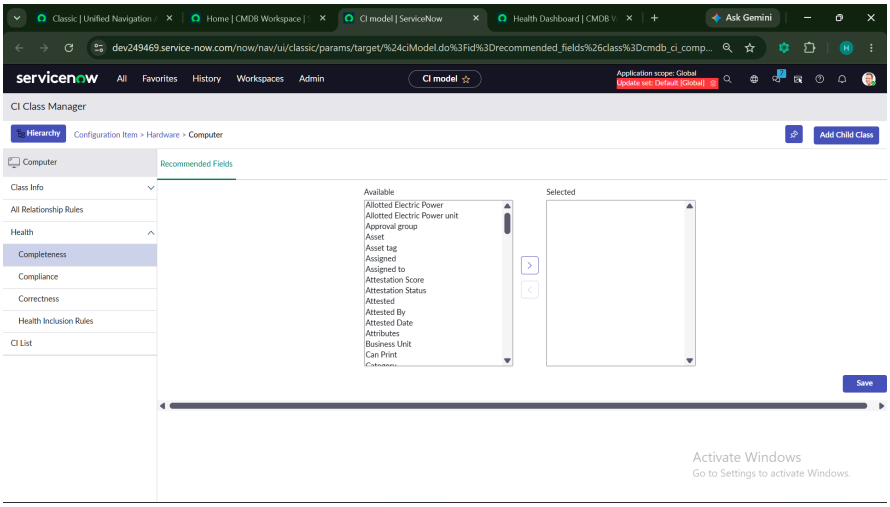
Answer : Inside CI Class Manager > Health

These rules are set **per CI class**. Rules set on a parent class also apply to its children.

a. **Completeness :** Which fields must be filled

“Every computer must have an owner”

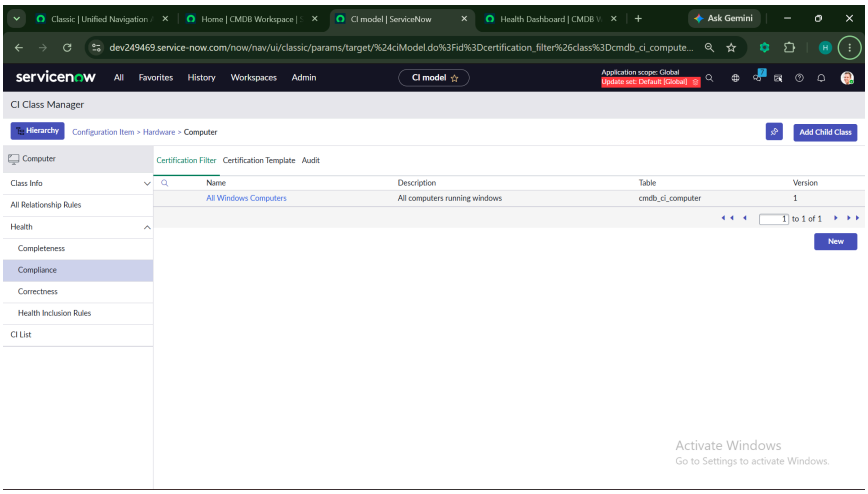
1. Open **Completeness**.
2. Move **Assigned to** from *Available* to *Selected*.
3. Click **Save**.



b. Compliance: All three work together to check compliance:

- Certification Filter: picks which CIs to check. Example: *All Windows Computers*.
- Certification Template: the rules to check on those CIs. Example: *OS must not be Windows XP*, *Owner must be filled*.
- Audit: runs the filter and template on a schedule, then shows which CIs pass or fail.

Flow: Filter (which CIs) → Template (what to check) → Audit (run and get results).



c. Correctness : When a CI is stale or orphan

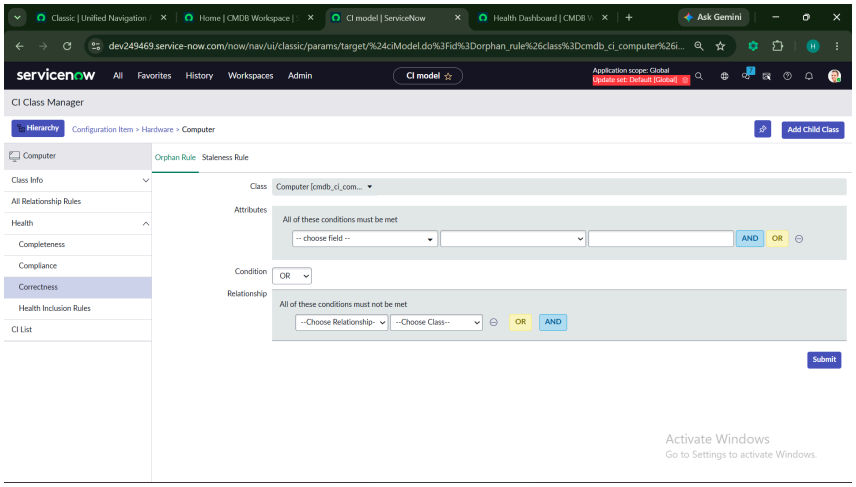
Orphan Rule defines when a **Computer** CI counts as an **orphan**: a CI that is lost or disconnected, with no useful data or links.

On your screen:

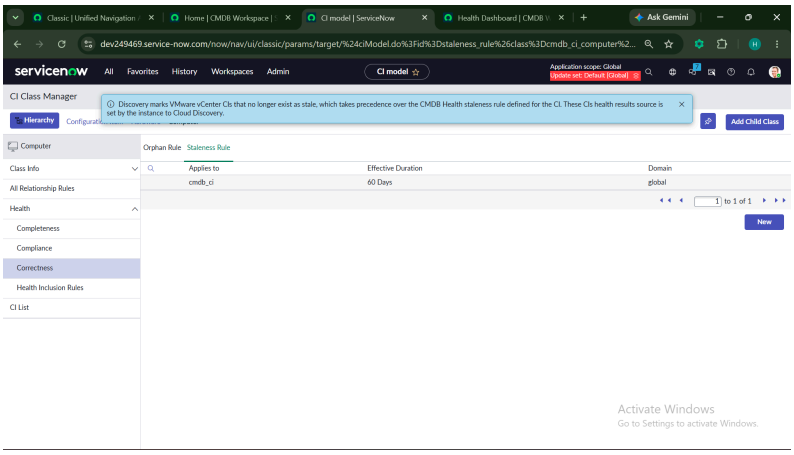
- **Attributes**: fields that must be empty or missing (for example, *no Assigned to*).
- **Condition (OR/AND)**: whether one or all of the checks must match.
- **Relationship**: links the CI must not have (for example, *not Runs on any Application*).

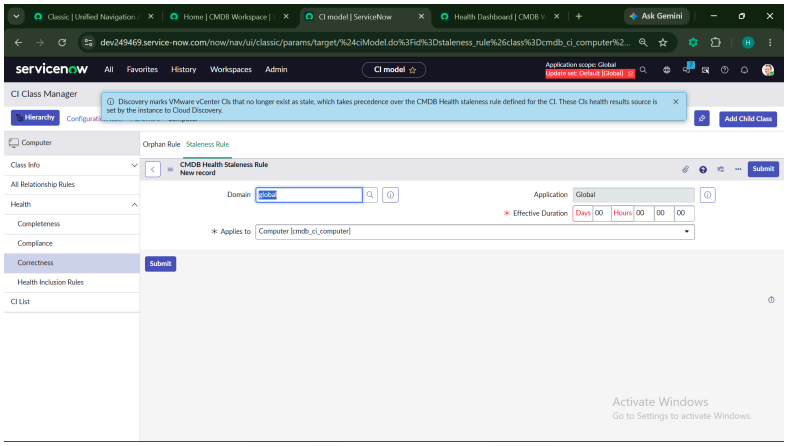
Example: "A computer with no **Assigned to** OR no **Runs on** relationship is an orphan."

Click **Submit** to save the rule.



For stale :



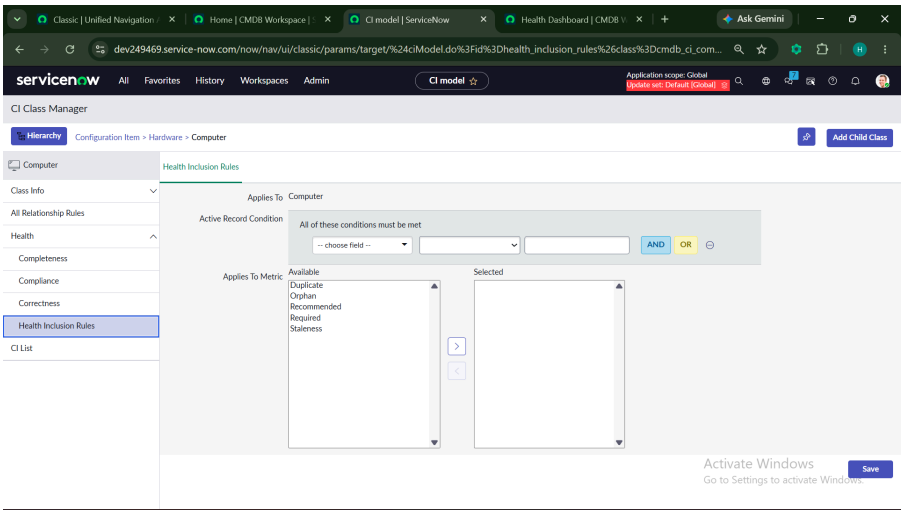


d. Health Inclusion Rule: It decides which Computer CIs are included in the health checks.

- Active Record Condition: the filter. Only CIs matching it are checked. Example: *Install status = Installed*, so retired computers are skipped.
- Applies To Metric: which checks use this filter: Duplicate, Orphan, Recommended, Required, Staleness. Move them to *Selected*.

Example: Only check Installed computers for Staleness and Orphan. Retired ones won't lower your score.

Click Save after setting it**.**



7. Relationship Health Dashboard: It checks the quality of the links between CIs (like Depends on, Runs on, Hosted on) and shows where they are duplicate, orphan, stale, or break rules.

Top 3 donut charts (238 relationships checked)

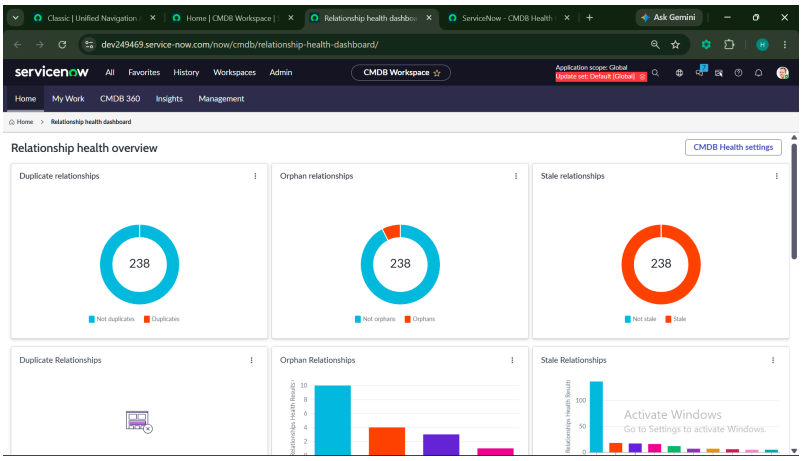
- **Duplicate relationships**: the same link saved twice. All blue, so none.
- **Orphan relationships**: links where one side is missing or the CI is gone. A small orange part means a few are orphans.
- **Stale relationships**: links not updated for a long time. All orange, so **all 238 are stale**.

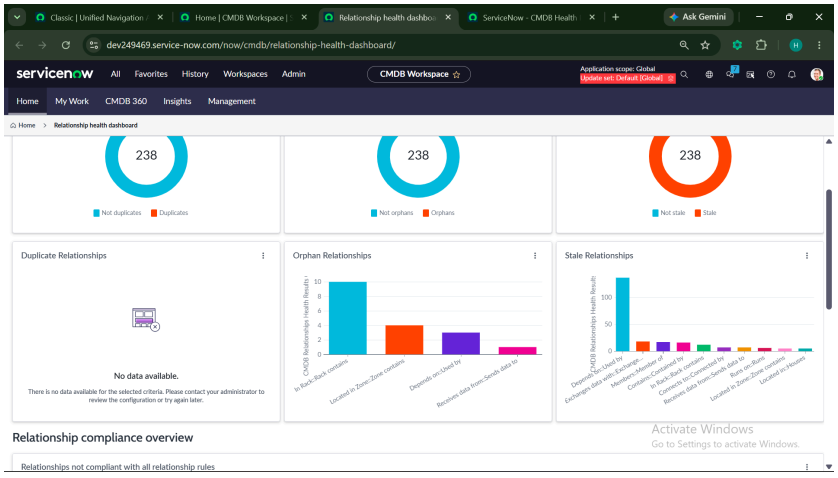
Bar charts below them

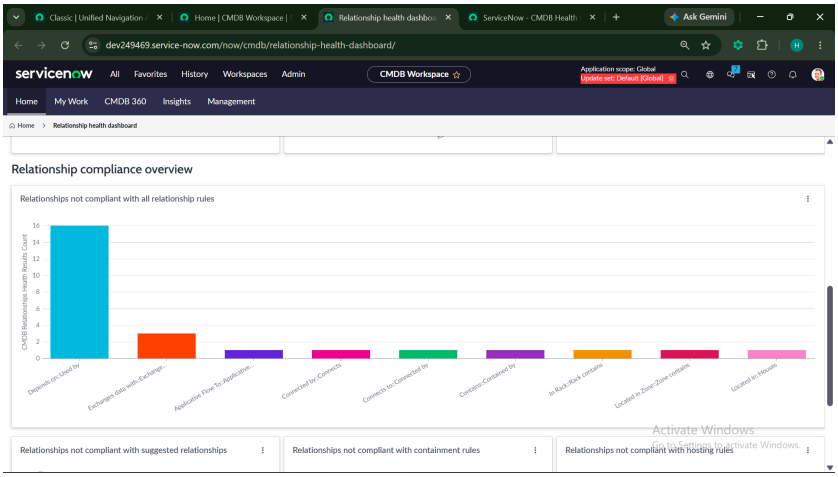
- Show the same problems **by relationship type**, for example *Depends on::Used by*, *Located in Zone*.
- Duplicate has "No data" because no duplicates exist.

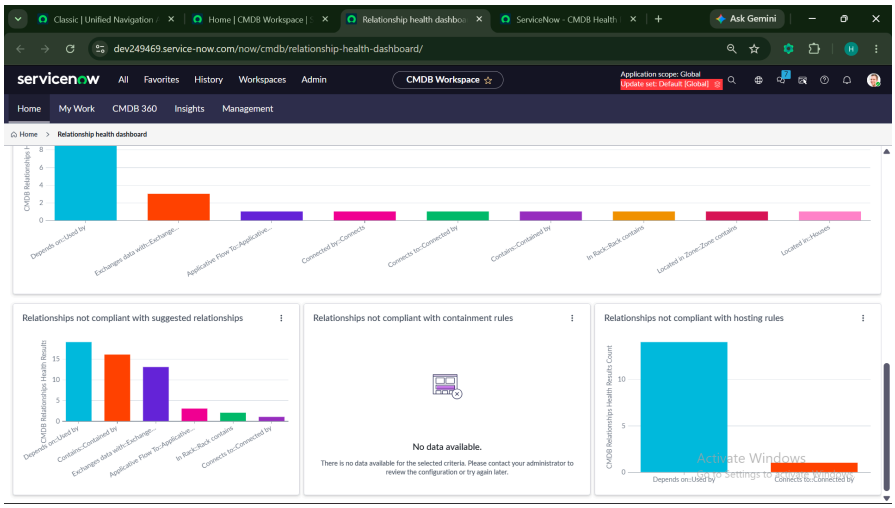
Relationship compliance overview

- **Not compliant with all relationship rules**: links that break any rule. Most are *Depends on::Used by* (about 16).
- **Not compliant with suggested relationships**: links that don't match the recommended pattern (for example, an app should *Run on* a server).
- **Not compliant with containment rules**: wrong container links (for example, a rack containing the wrong item). No data, so none.
- **Not compliant with hosting rules**: wrong hosting links (for example, an app hosted on the wrong CI type). A few exist.



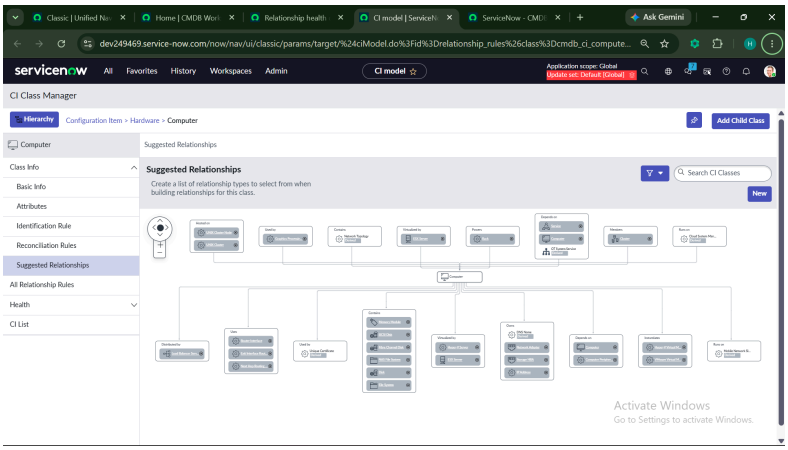

Classic | Unified Navigation
Home | CMDB Workspace |
Relationship health dashbo
ServiceNow - CMDB Health
Ask Gemini
dev249469.service-now.com/now/cmdb/relationship-health-dashboard/
servicenow
All
Favorites
History
Workspaces
Admin
CMDB Workspace
Application scope: Global
Update set: Default [Global]
Home
My Work
CMDB 360
Insights
Management
Home > Relationship health dashboard
238
Not duplicates
Duplicates
238
Not orphans
Orphans
238
Not stale
Stale
Duplicate Relationships
No data available.
There is no data available for the selected criteria. Please contact your administrator to review the configuration or try again later.
Orphan Relationships
CMDB Relationships Health Results Count
In Rack::Rack contains
Located in Zone::Zone contains
Depends on::Used by
Receives data from::Sends data to
Stale Relationships
Depends on::Used by
Exchanges data with::Exchange...
Member of::Exchange...
Contains::Contained by
In Rack::Rack contains
Connects to::Connected by
Receives data from::Sends data to
Runs on::Runs
Located in Zone::Zone contains
Located in::Houses
Relationship compliance overview
Relationships not compliant with all relationship rules
Activate Windows
Go to Settings to activate Windows.

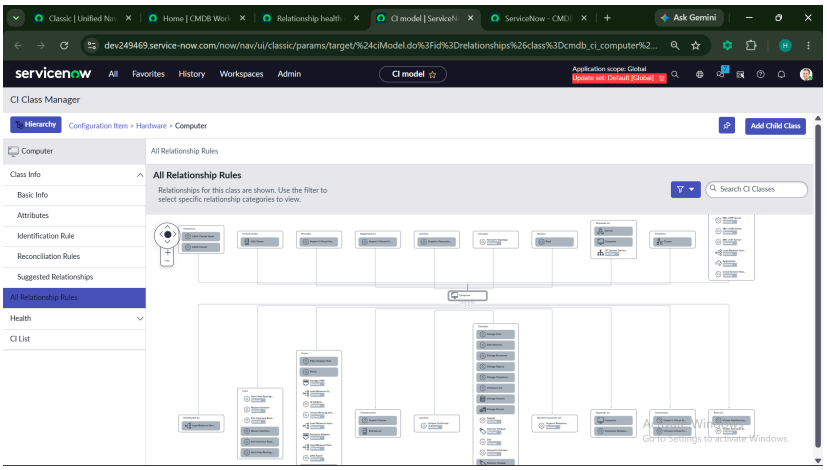

Classic | Unified Navigation
Home | CMDB Workspace |
Relationship health dashbo
ServiceNow - CMDB Health
Ask Gemini
dev249469.service-now.com/now/cmdb/relationship-health-dashboard/
servicenow
All
Favorites
History
Workspaces
Admin
CMDB Workspace
Application scope: Global
Update set: Default [Global]
Home
My Work
CMDB 360
Insights
Management
Home > Relationship health dashboard
Relationship compliance overview
Relationships not compliant with all relationship rules
CMDB Relationships Health Results Count
Depends on::Used by
Exchanges data with::Exchange...
Application Flow To::Applicative...
Connected by::Connects
Connects to::Connected by
Contains::Contained by
In Rack::Rack contains
Located in Zone::Zone contains
Located in::Houses
Activate Windows
Go to Settings to activate Windows.
Relationships not compliant with suggested relationships
Relationships not compliant with containment rules
Relationships not compliant with hosting rules

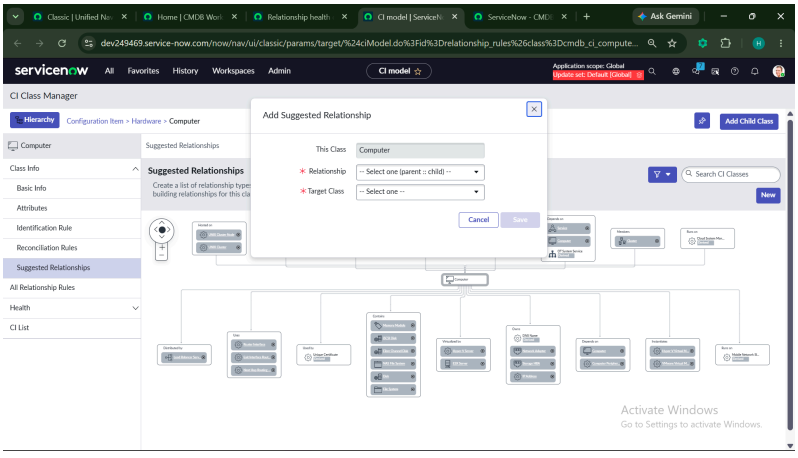

Classic | Unified Navigation
Home | CMDB Workspace |
Relationship health dashbo
ServiceNow - CMDB Health
Ask Gemini
dev249469.service-now.com/now/cmdb/relationship-health-dashboard/
servicenow
All
Favorites
History
Workspaces
Admin
CMDB Workspace
Application scope: Global
Update set: Default [Global]
Home
My Work
CMDB 360
Insights
Management
Home > Relationship health dashboard
CMDB Relationships Health Results Count
Depends on::Used by
Exchanges data with::Exchange...
Application Flow To::Applicative...
Connected by::Connects
Connects to::Connected by
Contains::Contained by
In Rack::Rack contains
Located in Zone::Zone contains
Located in::Houses
Relationships not compliant with suggested relationships
Relationships not compliant with containment rules
No data available.
There is no data available for the selected criteria. Please contact your administrator to review the configuration or try again later.
Relationships not compliant with hosting rules
Activate Windows
Go to Settings to activate Windows.

Question : Similarly, like a health dashboard, where can I configure my own relationship rules ?

Answer : In CI Class Manager, pick a class, then open All Relationship Rules







These are **relationship diagrams for the Computer class**, not a rule editor.

- **Suggested Relationships**: the types you *can pick* when linking a Computer (for example, Runs on, Depends on, Hosted on). You can add your own with **New** (top right).
- **All Relationship Rules**: shows every relationship the Computer class has, including inherited ones (read-only view). Use the **filter** to see one category.

To set your own rule: open **Suggested Relationships**, click **New**, and pick the relationship type and the related CI class (for example, Computer *Depends on* Service).

Containment, hosting and dependency rules are system-defined, so you can't edit them here.

8. Create CI: This is the Create a new CI wizard. It adds one CI manually, using IRE (Identification and Reconciliation Engine), which checks for duplicates before saving.

Steps (left side)

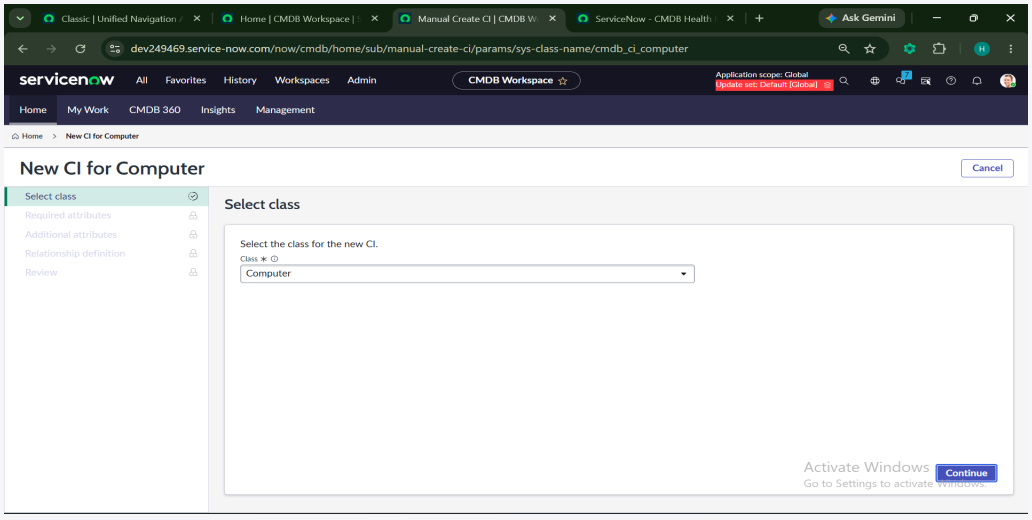
1. Select class: choose the CI type (Server, Computer, Application, etc.).
2. Required attributes: fill the must-have fields (for example, Name).
3. Additional attributes: fill optional fields (Owner, Location, OS).
4. Relationship definition: link the new CI to others (for example, *Runs on* a server).
5. Review: check everything, then submit.

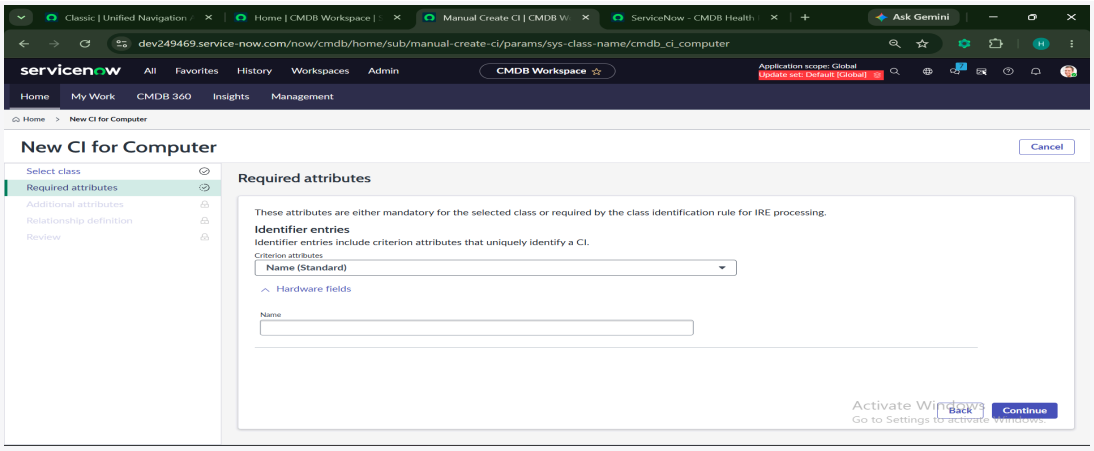
The later steps are locked until you finish the current one.

Example: Select Windows Server → enter name *SAP-WEB05* → add Owner and Location → link it to *SAP Financial Accounting* → Review → Submit.

Why IRE matters: if a CI with the same identifying fields already exists, it updates that CI instead of creating a duplicate.

Start by choosing a class, then click Continue.





“Hardware fields” is just the **group heading** for the field. The **Name** field belongs to the **Hardware** class, which is the parent of **Computer** (Configuration Item > Hardware > Computer).

You chose **Name (Standard)** as the identifier, so the form shows the **Name** box. It is placed under the class where that field is defined.

Type a name (for example, *SAP-WEB05*) and click **Continue**.

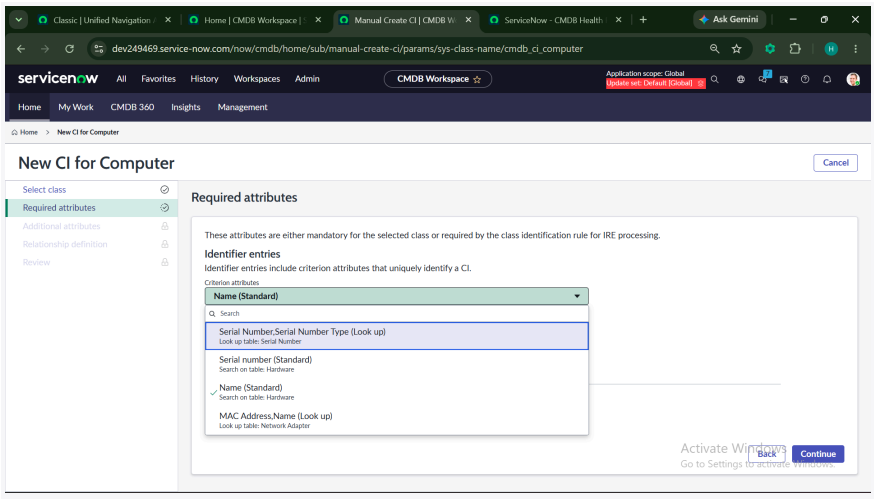
The name **SAP-WEB05** is what IRE uses to **identify** the CI.

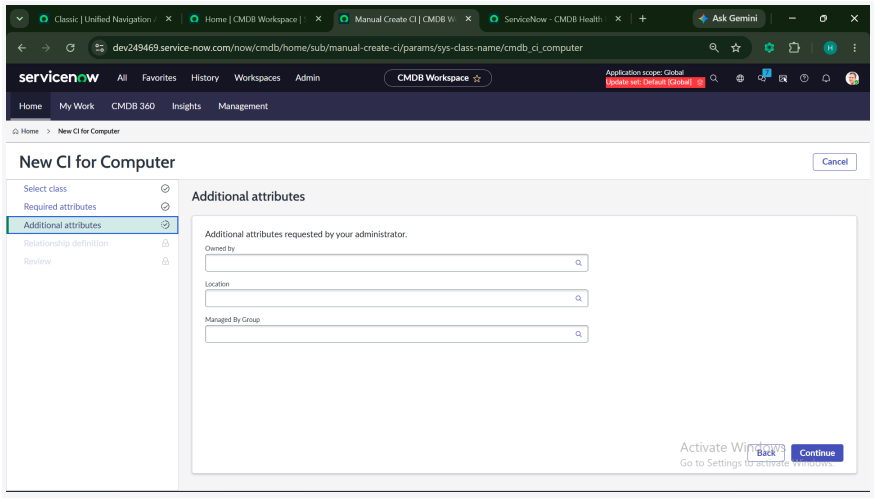
What happens:

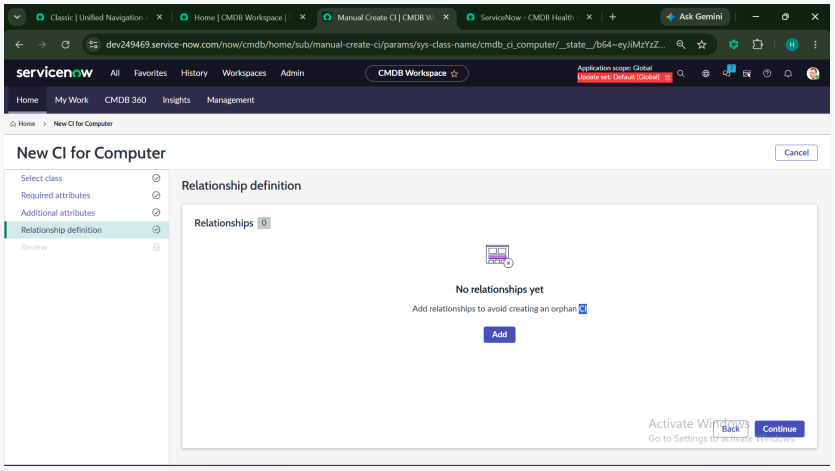
- IRE searches the CMDB for a Computer with the same name.
- **Found**: it updates that CI, so no duplicate is created.
- **Not found**: it creates a new CI.

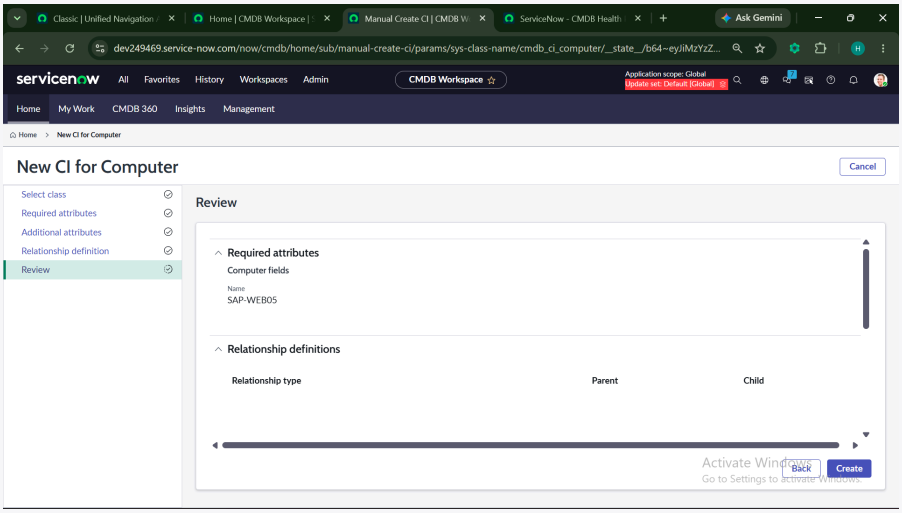
Why it asks now: Name is the **identifier** for the Computer class, so IRE can't check for duplicates without it. That is why it is on the **Required attributes** step, before the optional ones.

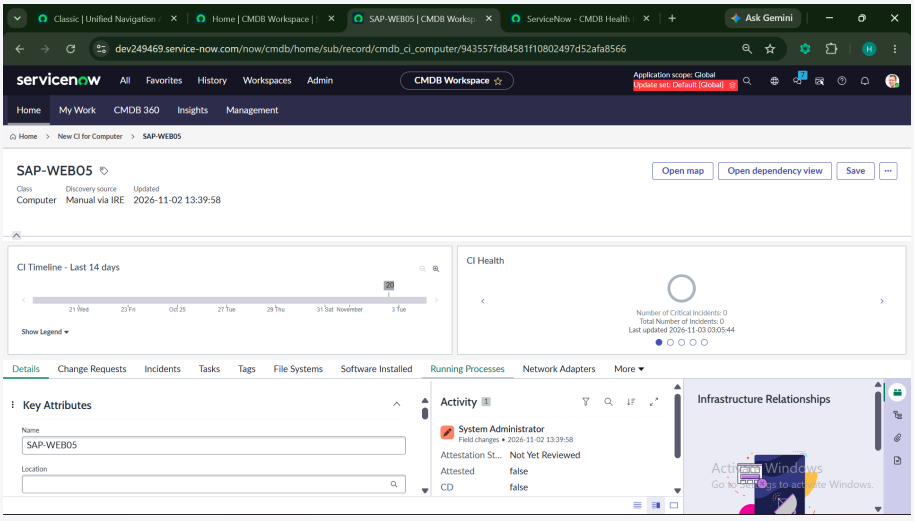
And notice here i can only chose those fields which are set as identifier in IRE:







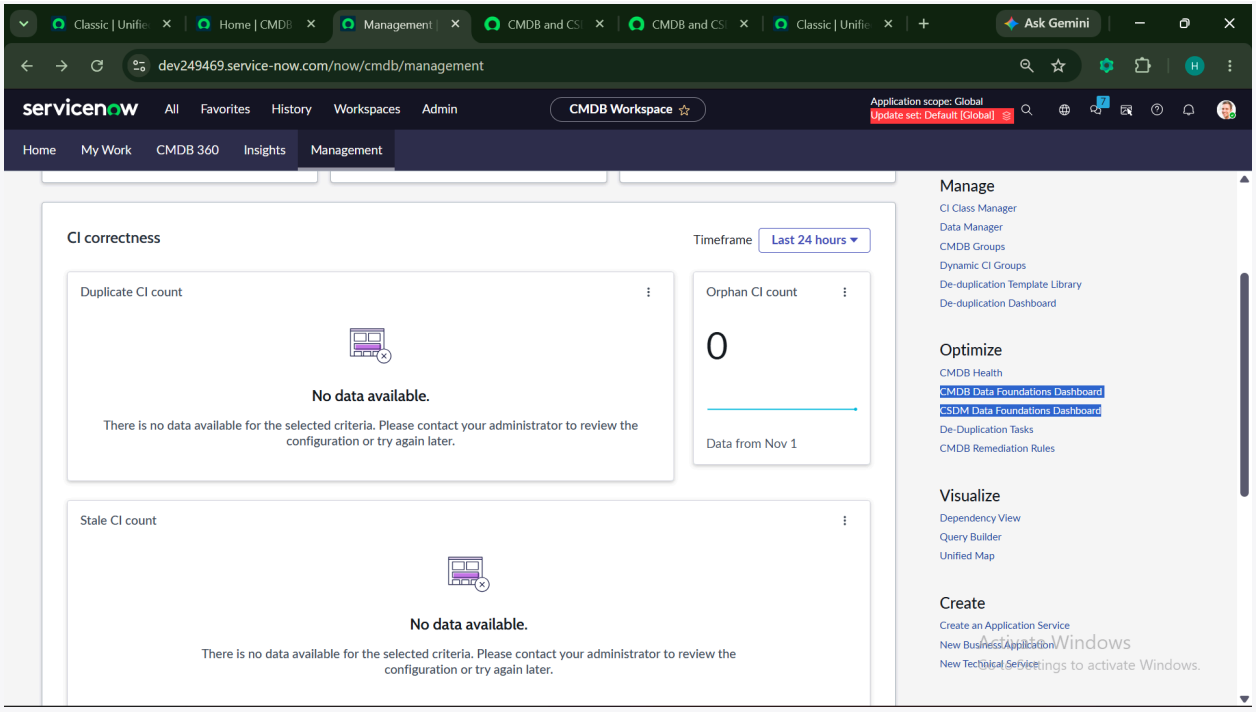


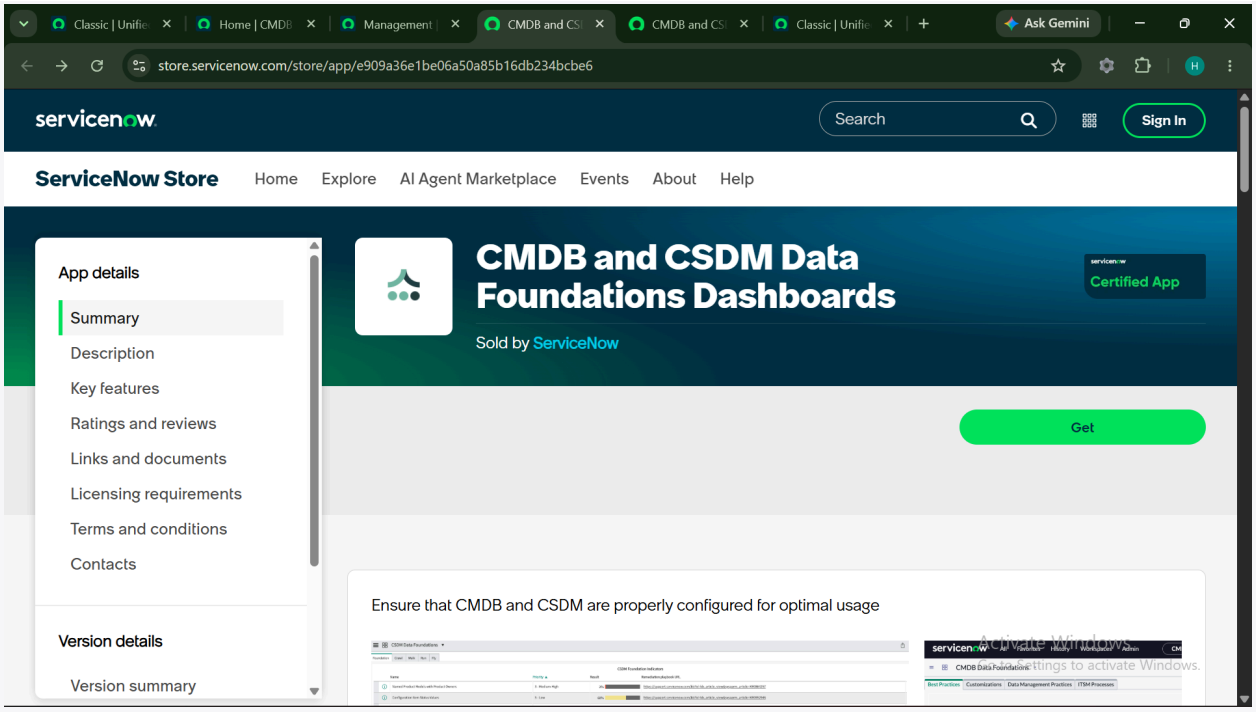


CI is created!!!

SOME OTHER IMPORTANT LINKS AND CONCEPTS :

1.





Both links (CMDB Data Foundations Dashboard and CSDM Data Foundations Dashboard) redirect to the same ServiceNow Store page because they are packaged

and installed together in a single application store app named CMDB and CSDM Data Foundations Dashboards.

What each link is meant to check:

1. CMDB Data Foundations Dashboard: Checks if your hardware, servers, and devices (CIs) follow basic setup rules (e.g., checking if servers are missing serial numbers, names, or owners).
2.
3. CSDM Data Foundations Dashboard: Checks if your business services & applications are correctly mapped to that hardware (e.g., checking if a payment app is properly linked to its server).
4.

Easy Summary:
They are two sides of the same plugin—one checks your hardware data health, and the other checks your service mapping health. ServiceNow opens the Store page so you can install or update the app package that contains both dashboards.

2. **De-Duplication** means finding and removing **duplicate entries** of the same item so your database stays clean.

Example:

If Discovery scans your network and accidentally creates two separate records for the exact same laptop:

- Record 1: Laptop-101 (IP: 192.168.1.50)
-
- Record 2: Laptop-101 (IP: 192.168.1.50)
-

De-Duplication detects that both records refer to the same physical device and merges or deletes the duplicate.

Here is how the two work together:

- **IRE (Identification and Reconciliation Engine):** Acts as the **preventative filter** for *incoming* CIs. When Discovery, SCCM, or an API tries to insert a new CI, IRE checks existing rules to prevent creating a duplicate in the first place.
- **De-Duplication (Data Manager / Tasks):** Handles **existing duplicates** already sitting in your database. It finds CIs that slipped past rules in the past (or were created before IRE rules were set up) and helps you merge or clean them up.

In short:

- **IRE** = Stops *future* duplicates from being created.
- **De-Duplication** = Cleans up *past* duplicates that already exist.

3. How do we handle deduplication?

In ServiceNow, handling duplicates involves identifying existing duplicates and then remediating (cleaning) them through automated rules or manual tasks.

Here is how the end-to-end process works:

Step 1: Identification (Finding the Duplicates)

ServiceNow runs background jobs that evaluate CIs against Identification Rules (e.g., matching Serial Number or MAC Address). When it finds CIs sharing the same identity details, it generates a De-duplication Task.

Step 2: Remediation (Resolving the Duplicates)

You can resolve duplicate tasks manually or automatically using Remediation Scripts or CMDB Data Manager:

1. Merge (Recommended): The primary CI is updated with the most recent attributes, related records (like Incidents/Changes) are re-linked to it, and the duplicate CI is retired/deleted.
2.
3. Delete / Retire: Deletes or marks the duplicate CI as retired.
4.
5. Ignore / Reclassify: Marks the task as non-duplicate if the CIs are actually separate physical assets.
6.

Real-World Example

The Problem:

SCCM and Discovery both imported the same server into the CMDB:

- CI 1: Server-01 (IP: 10.0.0.1, Serial: SN12345, created by SCCM)
-
- CI 2: Server-01 (IP: 10.0.0.1, Serial: SN12345, created by Discovery)
-

How ServiceNow Handles It:

Task Generation: ServiceNow creates a De-duplication Task (e.g., DUP0010023) pointing to both records.

Execution: An administrator or a automated Data Manager Policy opens the task:

- Selects CI 1 as the Master (Primary) CI.
-
- Merges relationships (e.g., moves 2 open Incidents from CI 2 over to CI 1).
-
- Automatically sets CI 2 state to Retired or deletes it.

4. how snow find deduplication task? is it by running IRE rules ?

Yes. ServiceNow creates De-Duplication tasks by applying **IRE (Identification and Reconciliation Engine)** rules.

Here is how it happens behind the scenes:

1. **Background Job Runs:** Scheduled jobs (like **CMDB Health Dashboard Correctness** jobs) scan your database periodically.
2.
3. **IRE Evaluates Rules:** The job runs existing CIs through the class **Identification Rules** (checking matching attributes like Serial Number, MAC Address, or Name).
4.
5. **Task Creation:** Whenever IRE detects two or more separate records matching the exact same identification criteria, it automatically generates a **De-Duplication Task** (DUPxxxxxxx) for you to review and clean up.

5. which schedule job is there to find out deduplication and create deduplication tasks? (IMPORTANT)

The primary scheduled job responsible for evaluating duplicate CIs and generating De-Duplication tasks is:

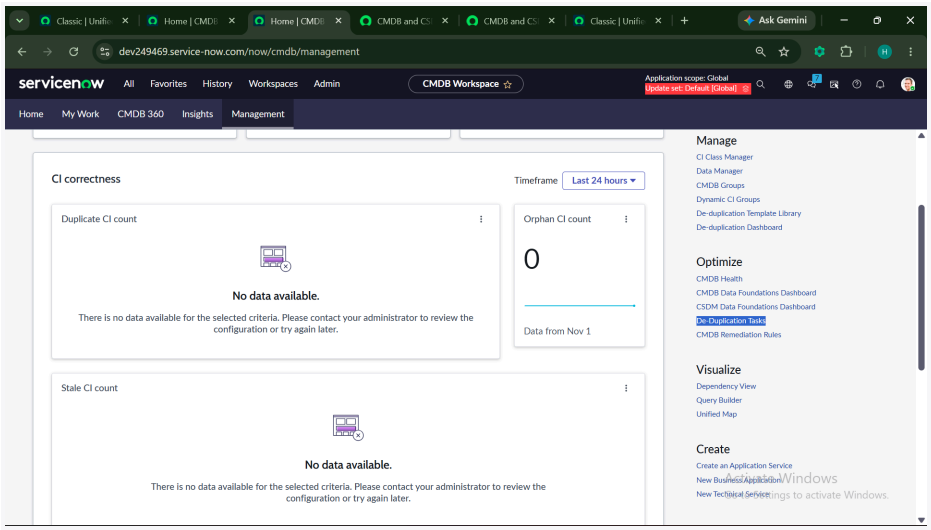
CMDB Health Dashboard - Correctness Score Calculation

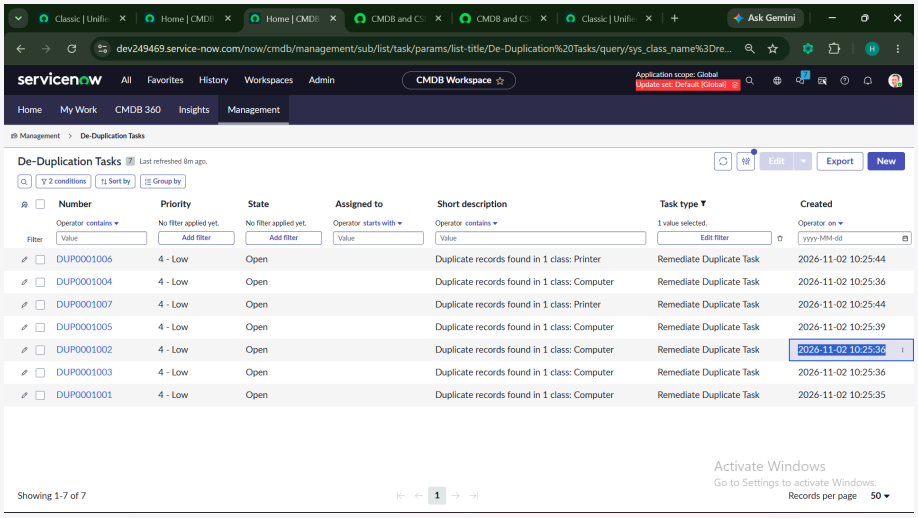
How it works:

1. When this scheduled job runs, it evaluates CIs against **CMDB Identification Rules (IRE)**.
2.
3. For any CIs matching the identification criteria, the Identification Engine flagged them as duplicates and automatically creates a record in the **De-Duplication Task (**reconcile_duplicate_task**)** table.
4.
5. Simultaneously, it updates the **Duplicate Metric** for the CMDB Health Dashboard.
6.

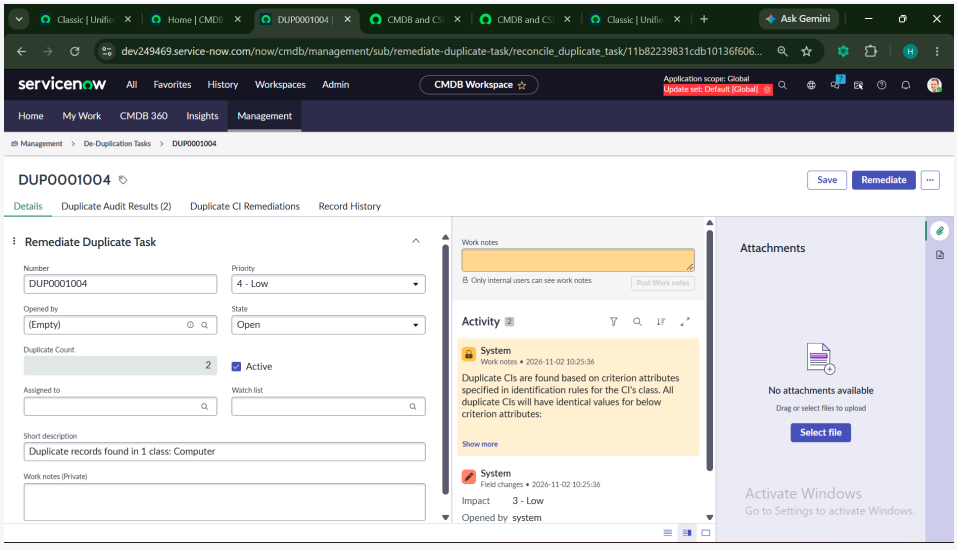
Verification: You can find and run this job anytime by going to **System Scheduler > Scheduled Jobs > Scheduled Jobs** (sys_trigger.list) and searching for CMDB Health Dashboard - Correctness Score Calculation.

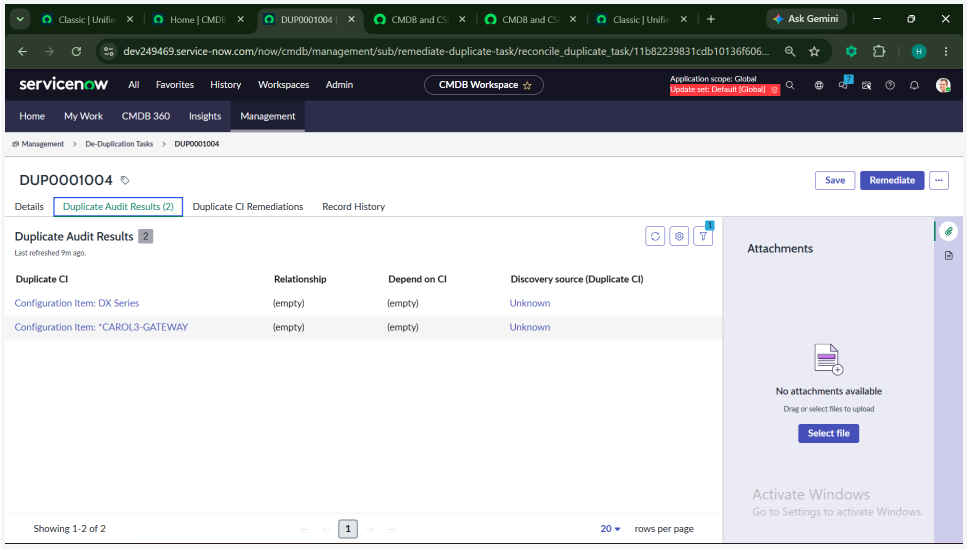
For a full step-by-step walkthrough of handling duplicate CIs and running remediation tasks, check out this guide on ServiceNow CMDB De-Duplication & Remediation. This video explains how identification rules trigger de-duplication tasks and demonstrates the remediation process in action.





As I ran the schedule job it detected these duplicate cis





Why they are listed: The system flagged these two computer records as representing the exact same physical asset based on matching IRE identification attributes.

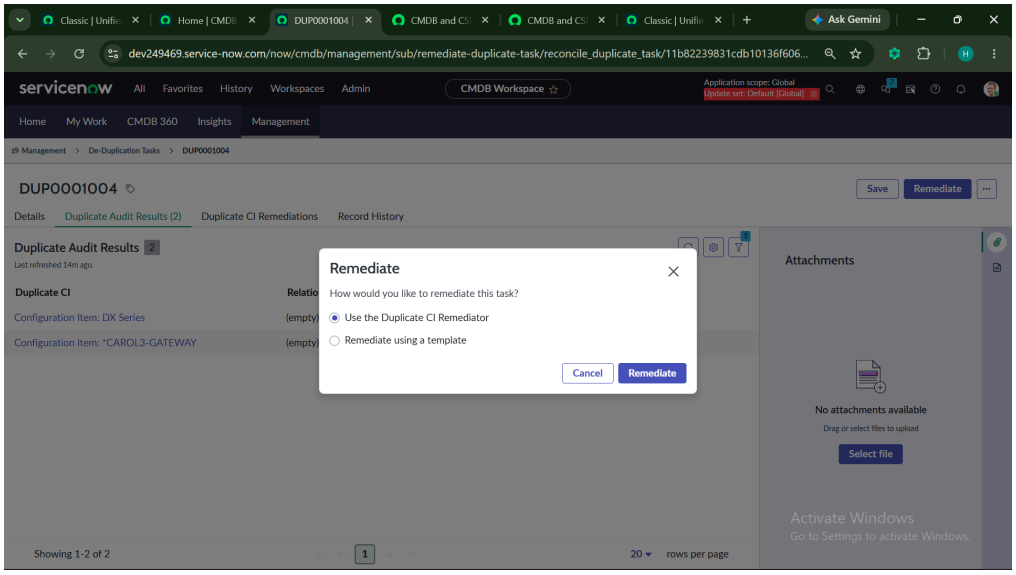
To resolve this duplicate task (DUP0001004), click the **Remediate** button in the top-right corner.

1.Click Remediate: Top-Right Corner.
Click the purple **Remediate** button at the top right of the form. *Verification:* A wizard popup titled **Remediate Duplicate Task** will open showing the 2 duplicate CIs side by side.

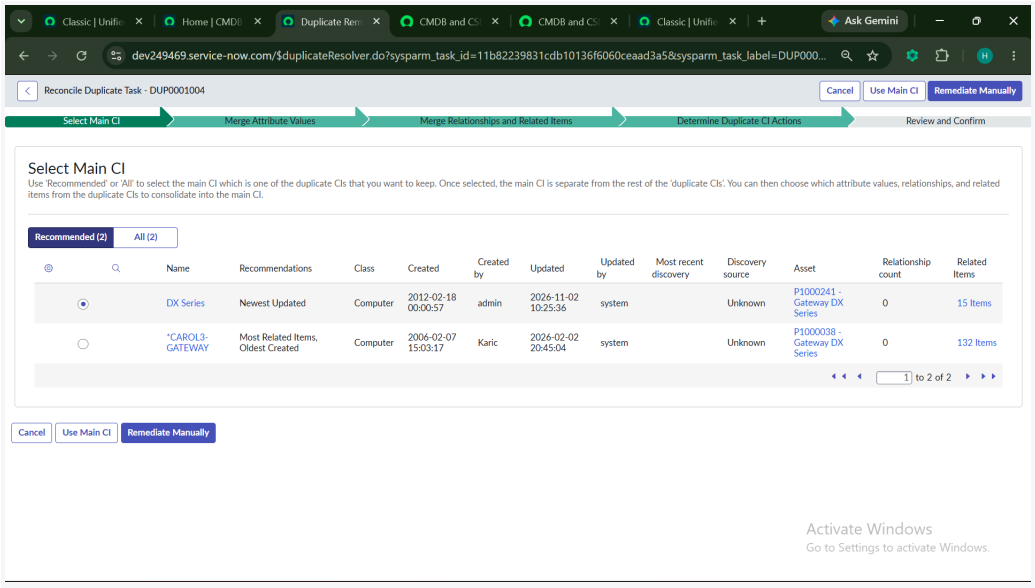
2.Select Master Record:
Choose which CI record to keep as the **Main (Master) CI** (usually the one with the most recent or complete data).

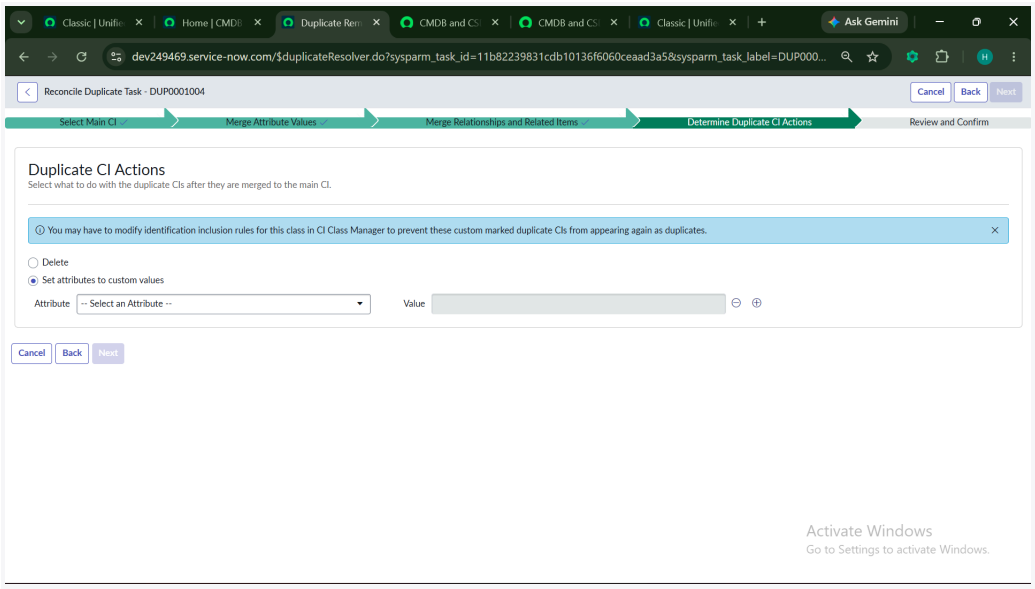
3.Choose Strategy & Finish:
Select what to do with the second record (**Merge** relationships or **Delete/Retire** the duplicate) and click **Next / Finish**.

Verification: The task state will update to **Closed Complete**, and the duplicate CI will be retired or deleted.



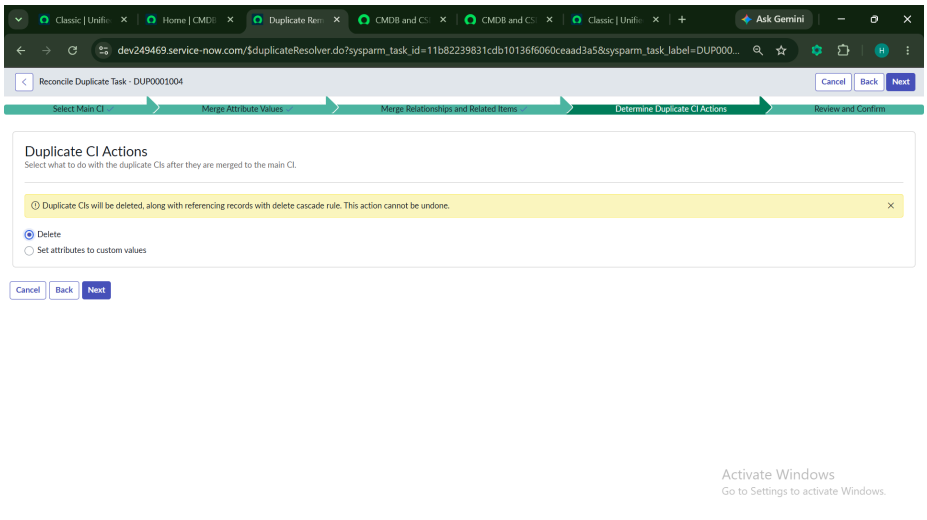
Use the Duplicate CI Remediator: Opens the interactive wizard that lets you compare the two records side-by-side, pick which one to keep as Master, and merge or delete the extra record.

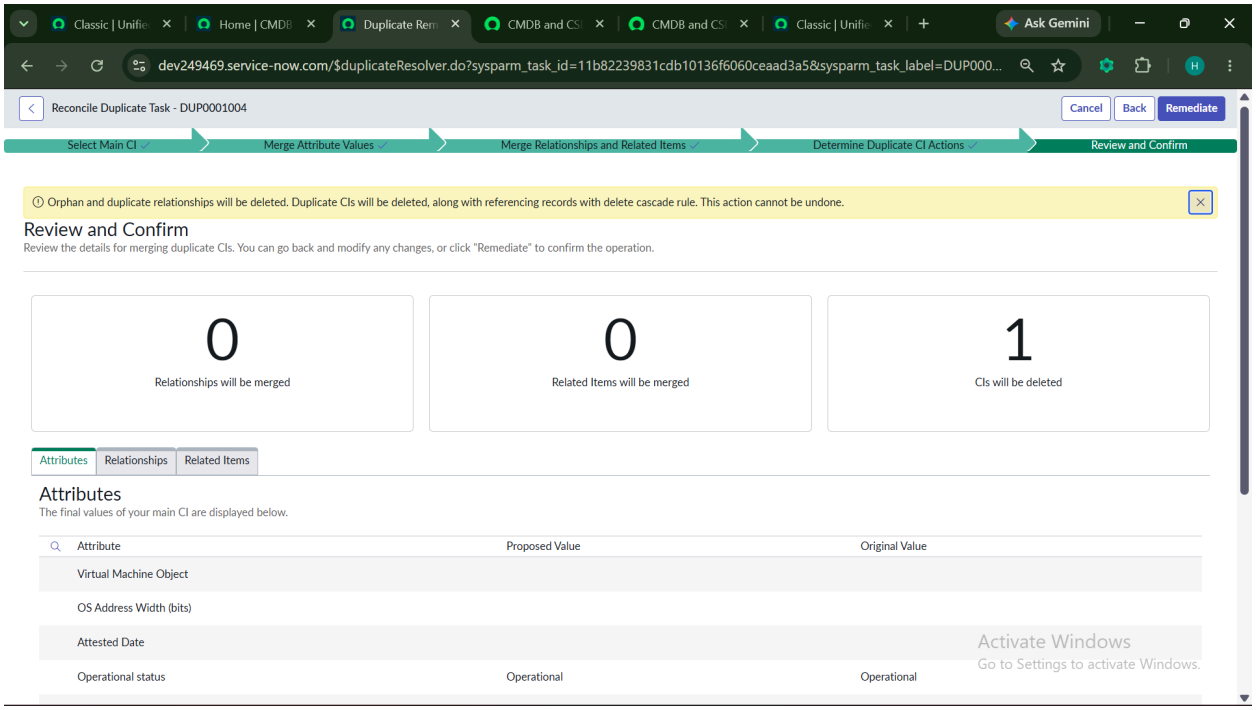




What to choose:

1. Option A: Select "Delete" (Recommended)
    - What it does: Completely deletes the duplicate record from the database so it stops showing up in searches.
    - Best when: You are sure the record is a useless copy.
2. Option B: Select "Set attributes to custom values" (Default)
    - What it does: Keeps the duplicate record in the system, but changes specific field values (e.g., setting Operational status to Retired or Install Status to Absent).
    - Best when: Your company requires an audit history of old records instead of hard-deleting them.



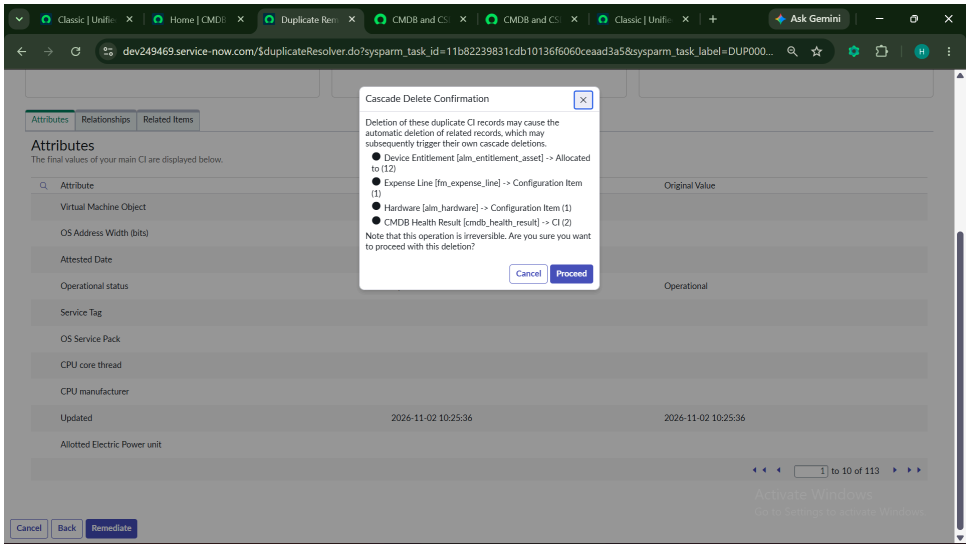


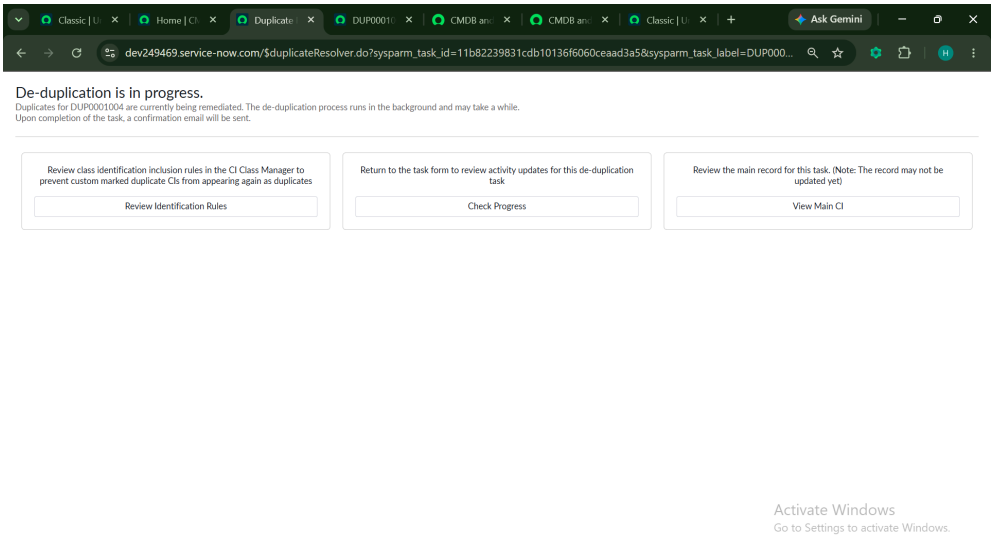
This is the final summary screen before ServiceNow executes the merge and cleanup.

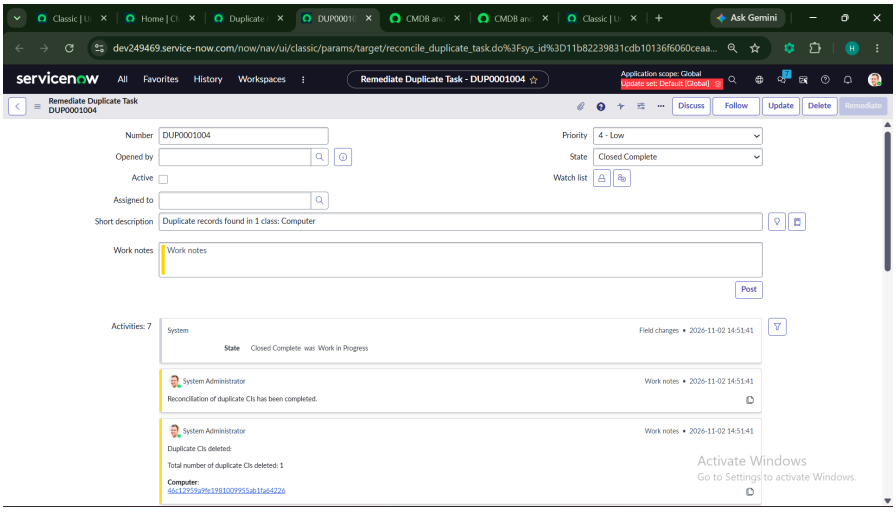
What this summary means:

- **0 Relationships / 0 Related Items to merge:** There are no active links or dependencies that need to be transferred to DX Series.
-
- **1 CI will be deleted:** The duplicate record (*CAROL3-GATEWAY) will be permanently removed from your system as you selected in the previous step.
-

1.Click Remediate to finish: Top-Right Corner.
Click the purple **Remediate** button in the top-right corner. *Verification:* The wizard will execute the merge, permanently delete the duplicate CI, and automatically close task DUP0001004 as **Closed Complete**.


Cascade Delete Confirmation
Deletion of these duplicate CI records may cause the automatic deletion of related records, which may subsequently trigger their own cascade deletions.
Device Entitlement [alm_entitlement_asset] -> Allocated to (12)
Expense Line [fm_expense_line] -> Configuration Item (1)
Hardware [alm_hardware] -> Configuration Item (1)
CMDB Health Result [cmdb_health_result] -> CI (2)
Note that this operation is irreversible. Are you sure you want to proceed with this deletion?
Cancel
Proceed
Attributes
Relationships
Related Items
Attributes
The final values of your main CI are displayed below.
Attribute
Original Value
Virtual Machine Object
OS Address Width (bits)
Attested Date
Operational status
Operational
Service Tag
OS Service Pack
CPU core thread
CPU manufacturer
Updated
2026-11-02 10:25:36
2026-11-02 10:25:36
Allotted Electric Power unit
1 to 10 of 113
Cancel
Back
Remediate


De-duplication is in progress.
Duplicates for DUP0001004 are currently being remediated. The de-duplication process runs in the background and may take a while.
Upon completion of the task, a confirmation email will be sent.
Review class identification inclusion rules in the CI Class Manager to prevent custom marked duplicate CIs from appearing again as duplicates
Review Identification Rules
Return to the task form to review activity updates for this de-duplication task
Check Progress
Review the main record for this task. (Note: The record may not be updated yet)
View Main CI

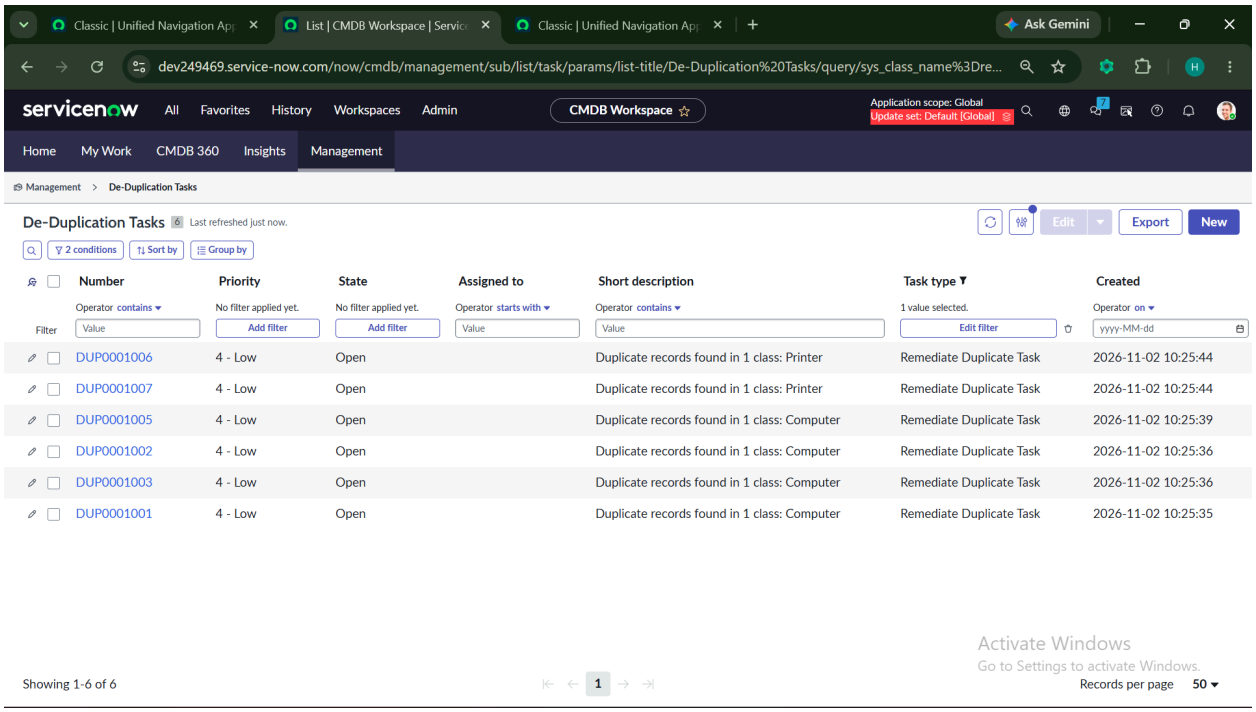

servicenow
Remediate Duplicate Task - DUP0001004
Remediate Duplicate Task
DUP0001004
Discuss
Follow
Update
Delete
Remediate
Number
DUP0001004
Priority
4 - Low
Opened by
State
Closed Complete
Active
Watch list
Assigned to
Short description
Duplicate records found in 1 class: Computer
Work notes
Post
Activities: 7
System
Field changes • 2026-11-02 14:51:41
State Closed Complete was Work in Progress
System Administrator
Work notes • 2026-11-02 14:51:41
Reconciliation of duplicate CIs has been completed.
System Administrator
Work notes • 2026-11-02 14:51:41
Duplicate CIs deleted:
Total number of duplicate CIs deleted: 1
Computer:

The task is officially finished and closed!

Key details on your screen:

- **State:** Closed Complete—the de-duplication job finished successfully.
-
- **Active:** Box is **unchecked**, meaning it's cleared from your active queue.
-
- **Activity Log:** Confirms that **1 duplicate Computer CI was deleted** and reconciliation is complete.
-

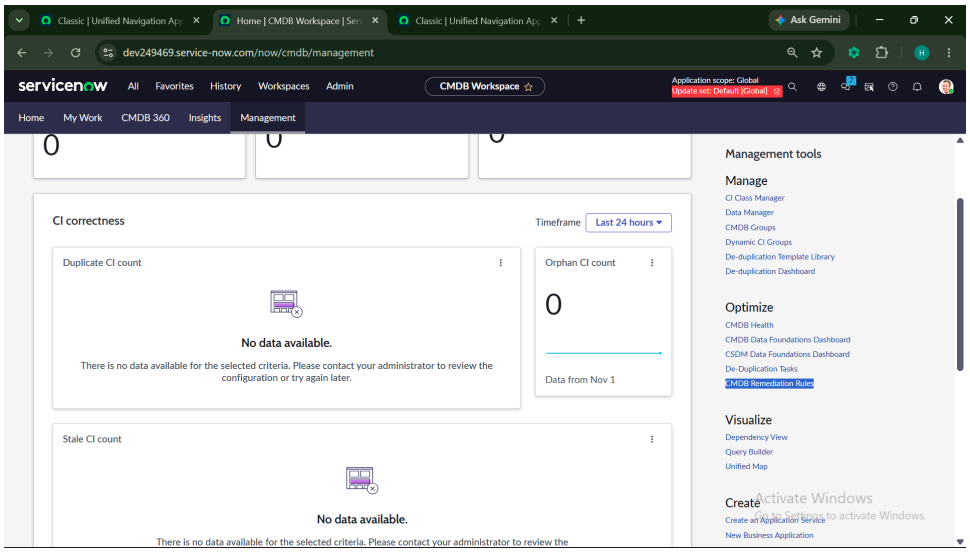
You can close this tab and go back to your main workspace—your active duplicate task count is now reduced!

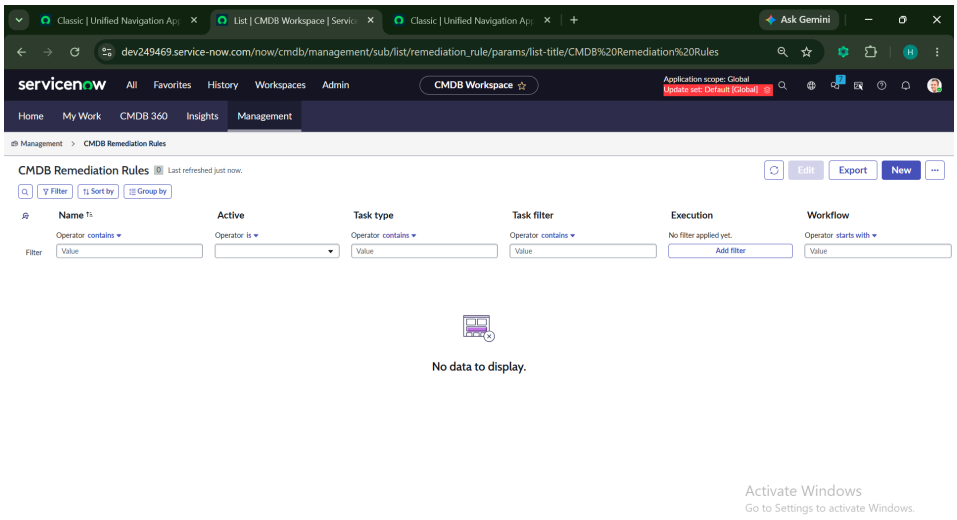


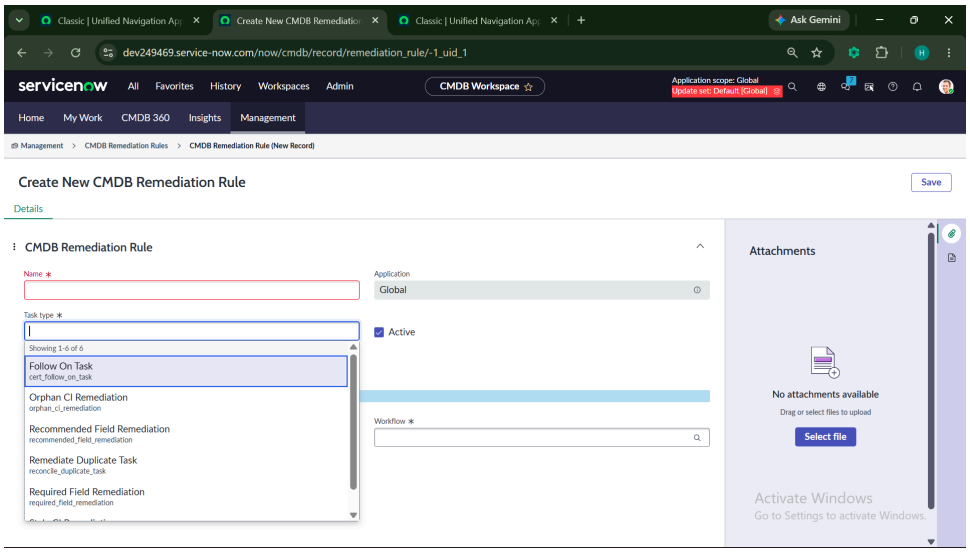
See that record is deleted now!

****

REMEDIATION RULES:


dev249469.service-now.com/now/cmdb/management
servicenow
CMDB Workspace
Home
My Work
CMDB 360
Insights
Management
CI correctness
Timeframe
Last 24 hours
Duplicate CI count
No data available.
There is no data available for the selected criteria. Please contact your administrator to review the configuration or try again later.
Orphan CI count
0
Data from Nov 1
Stale CI count
No data available.
Management tools
Manage
CI Class Manager
Data Manager
CMDB Groups
Dynamic CI Groups
De-duplication Template Library
De-duplication Dashboard
Optimize
CMDB Health
CMDB Data Foundations Dashboard
CSDM Data Foundations Dashboard
De-Duplication Tasks
CMDB Remediation Rules
Visualize
Dependency View
Query Builder
Unified Map
Create
Create an Application Service
New Business Application
Activate Windows


dev249469.service-now.com/now/cmdb/management/sub/list/remediation_rule/params/list-title/CMDB%20Remediation%20Rules
servicenow
CMDB Workspace
Home
My Work
CMDB 360
Insights
Management
Management > CMDB Remediation Rules
CMDB Remediation Rules
Edit
Export
New
Filter
Sort by
Group by
Name
Active
Task type
Task filter
Execution
Workflow
No filter applied yet.
Add filter
No data to display.
Activate Windows
Go to Settings to activate Windows.


dev249469.service-now.com/now/cmdb/record/remediation_rule/-1_uid_1
servicenow
CMDB Workspace
Home
My Work
CMDB 360
Insights
Management
Management > CMDB Remediation Rules > CMDB Remediation Rule (New Record)
Create New CMDB Remediation Rule
Save
Details
CMDB Remediation Rule
Name *
Application
Global
Task type *
Active
Showing 1-6 of 6
Follow On Task
cert_follow_on_task
Orphan CI Remediation
orphan_ci_remediation
Recommended Field Remediation
recommended_field_remediation
Remediate Duplicate Task
reconcile_duplicate_task
Required Field Remediation
required_field_remediation
Workflow *
Attachments
No attachments available
Drag or select files to upload
Select file
Activate Windows
Go to Settings to activate Windows.

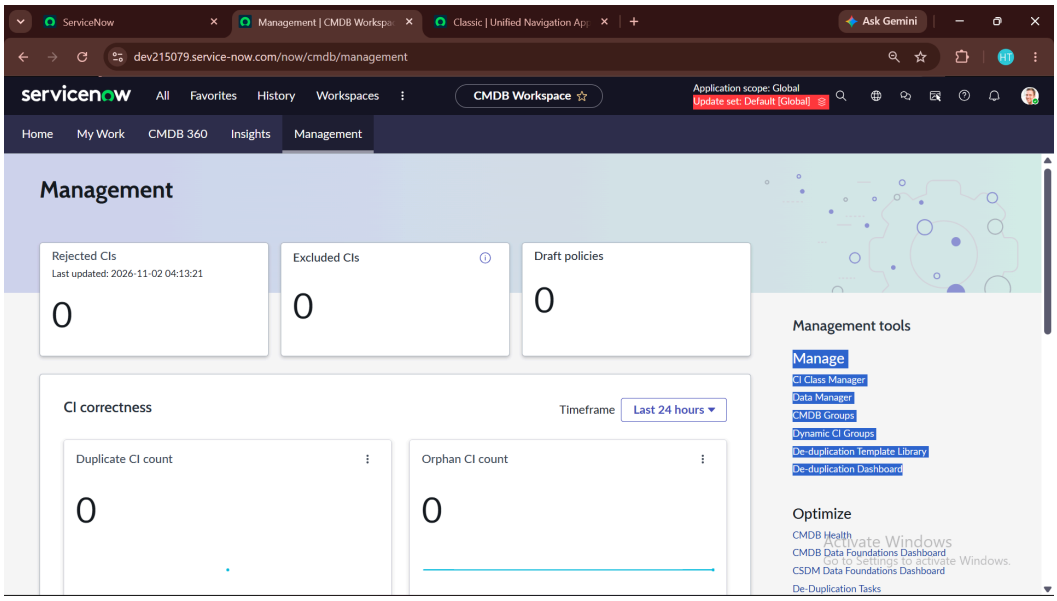
“CMDB Remediation Rules” are **“automated fix-it rules”** that tell ServiceNow how to fix health issues (like duplicates, stale CIs, or orphans) automatically without manual wizard clicks.

Easy Example:

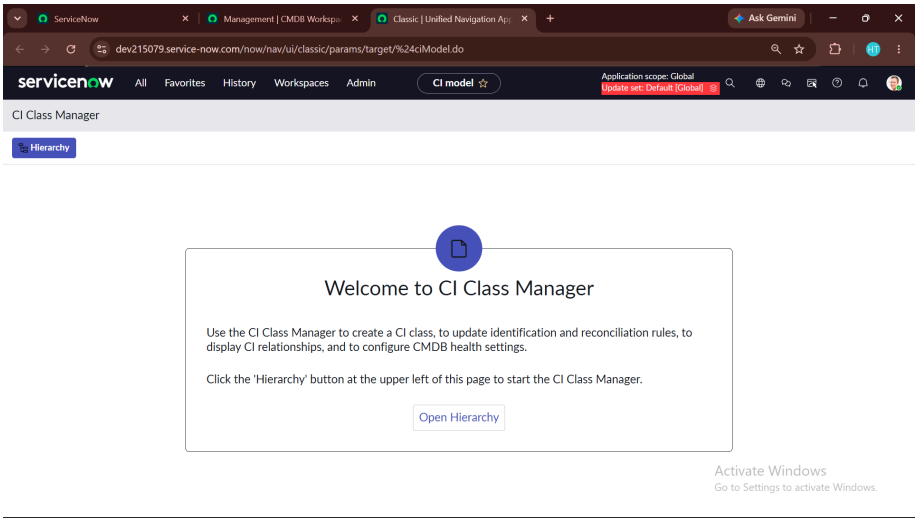
Instead of clicking through the remediation wizard manually for every duplicate task, you could create a rule here that says:

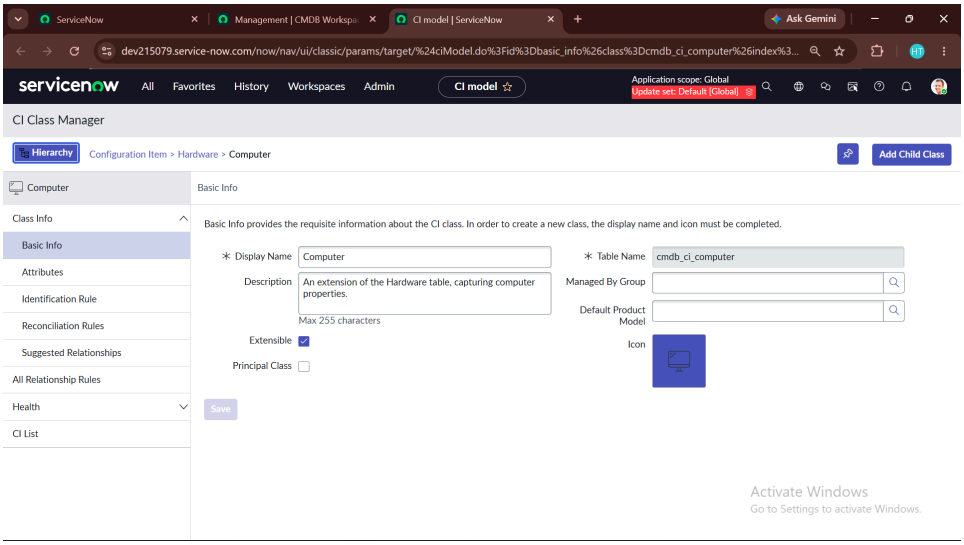
"Whenever a duplicate task is found for a Printer CI, automatically keep the newest record and delete the duplicate."

EXPLORING MANAGEMENT MODULES :



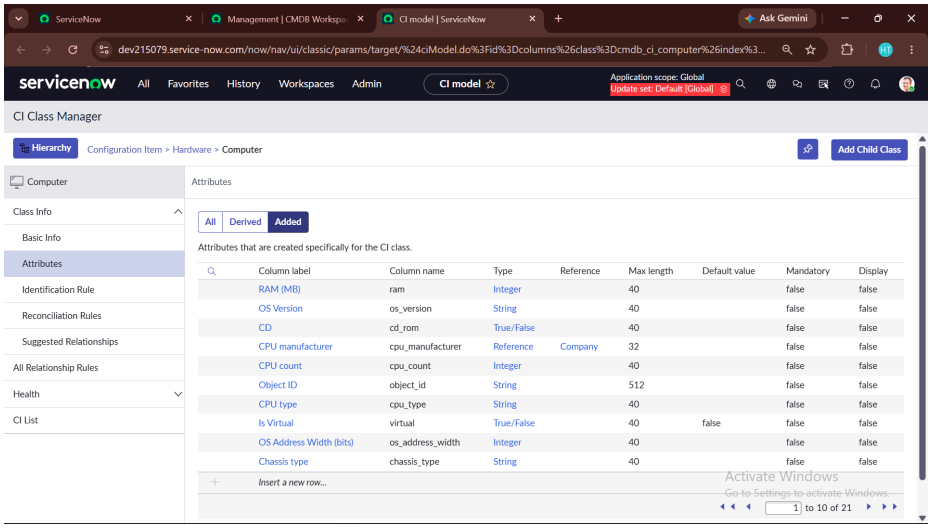
1. **CI Class Manager :** It is the ServiceNow tool to **create and manage CI classes** (types of CIs like Computer, Server, Application). It shows the class hierarchy and lets you set each class’s fields, rules, relationships and health settings.

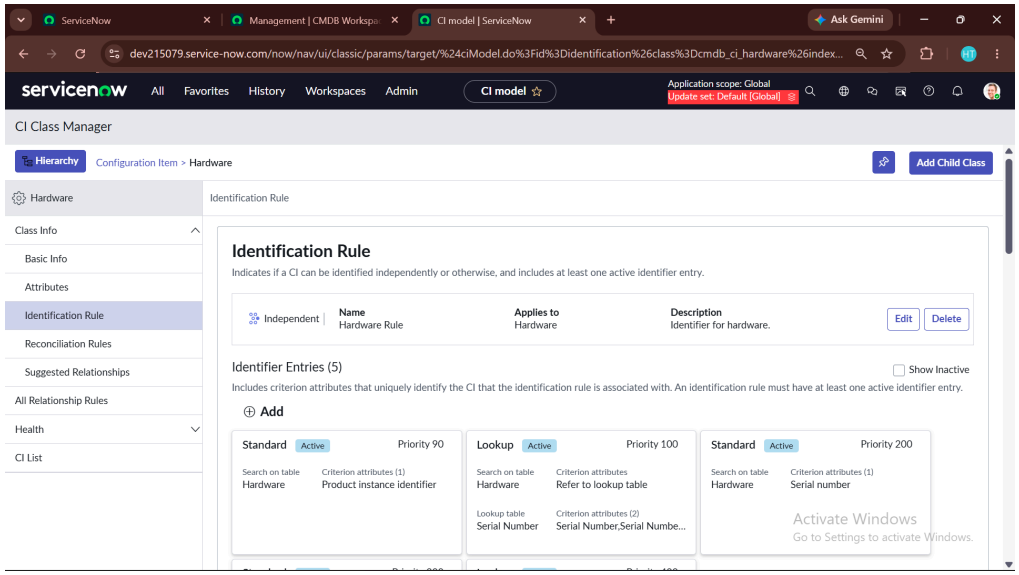




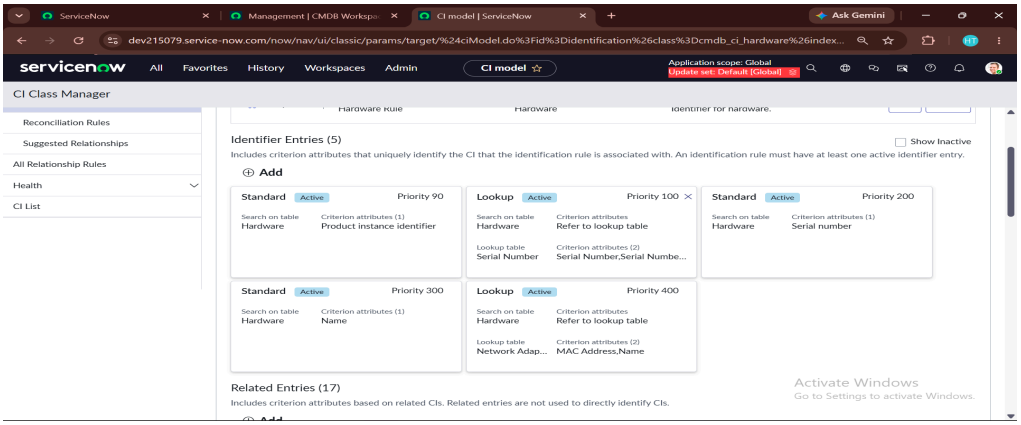
Default Product Model: the model (for example, *Lenovo 20BTS00R00*) that is filled in automatically when you create a CI of this class.

Principal Class: a checkbox marking this class as a **main class** used for software or service tracking. When ticked, CIs of this class can be treated as the principal (main) CI in Software Asset Management and discovery. For most custom classes, leave it unticked.

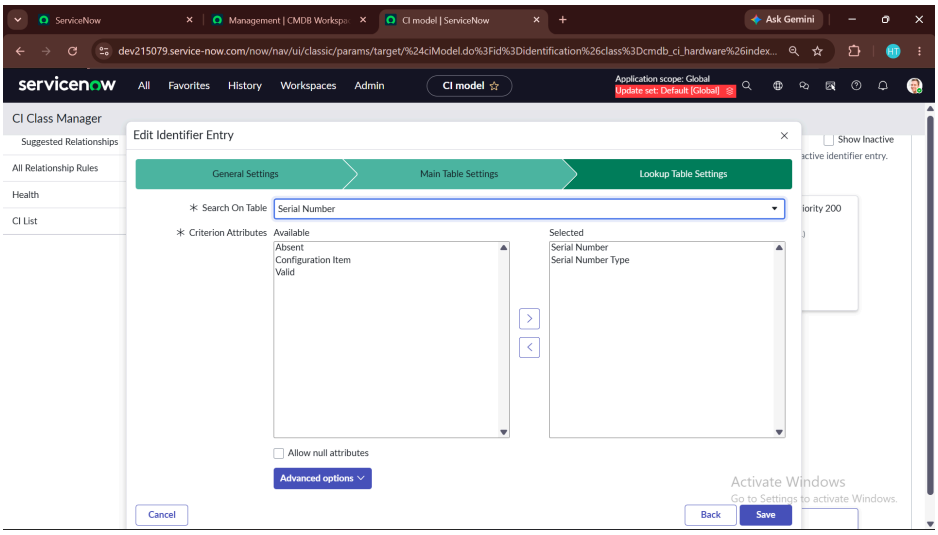




Identification Rule tells IRE how to recognize a Computer CI, so the same CI is never created twice.



- **Standard**: check fields on the **main table itself** (for example, *Name* on Hardware).
- **Lookup**: check fields on a **related table** that points back to the CI (for example, *Serial Number* table).

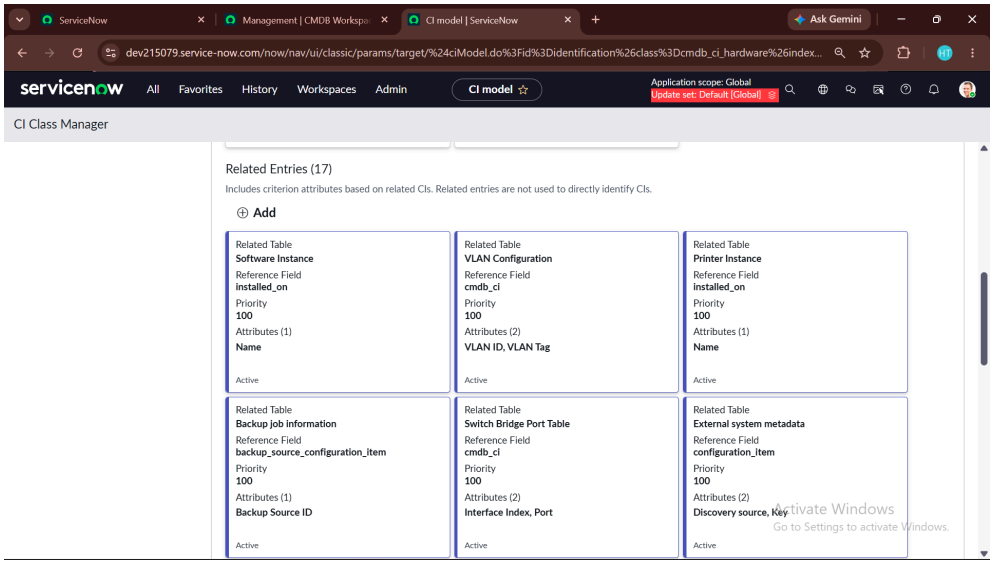


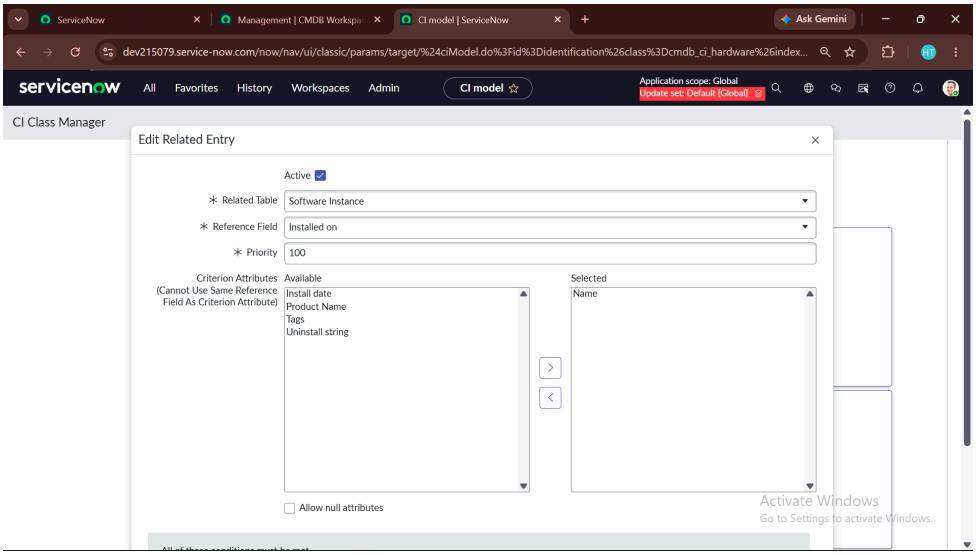
Doubt : what about all these classes ? why only these tables are come in hardware ?is there any reason behind it.

Solution : There is no special reason. These tables are **not linked to Hardware**.

The dropdown is just an **alphabetical list of many tables in the instance**. That's why you see unrelated ones like *Problem*, *Private Task* and *PostgreSQL Schema*. It is not filtered to Hardware.

Which one to pick: only a table that has a **reference field to a CI**, such as *Serial Number* or *Network Adapter*. Tables like *Problem* are not made for identifying CIs, so you would not use them.



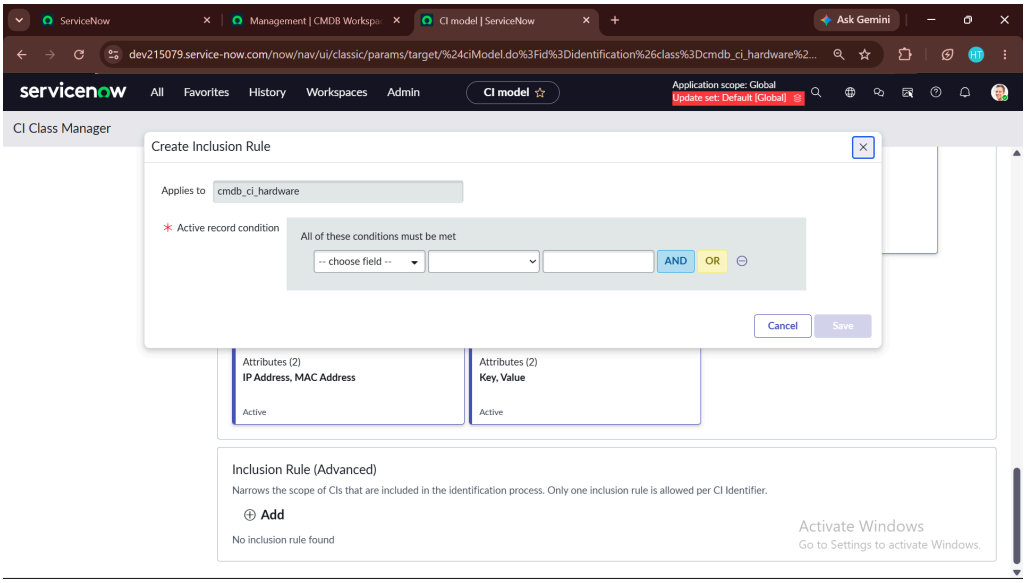


Related entries identify a dependent record using its own field plus its parent CI, and they also prevent duplicates.

Example (Software Instance):

- Rule: *Name + Installed on*
- Apache on **Laptop-01** exists. A new Apache on Laptop-01 arrives, so IRE updates it.
- Apache on **Laptop-02** arrives, so IRE creates a new record, because the parent is different.

Why: the name alone isn't unique, but name + parent hardware is.



Inclusion Rule (Advanced) limits **which CIs IRE uses for identification**. CIs that don't match the condition are ignored during matching.

Fields on the screen

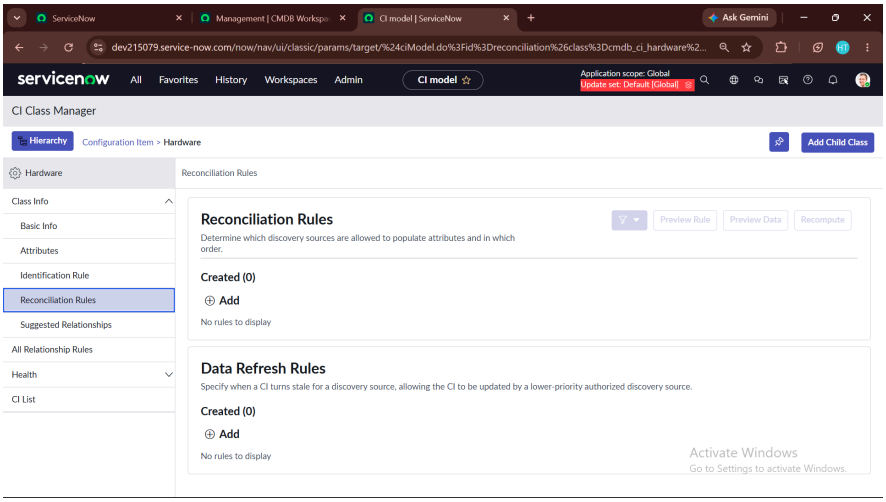
- **Applies to:** the class (cmdb_ci_hardware).
- **Active record condition:** the filter. Only CIs that match it are considered.
- **AND / OR:** combine several conditions.
- Only **one** inclusion rule is allowed per CI identifier.

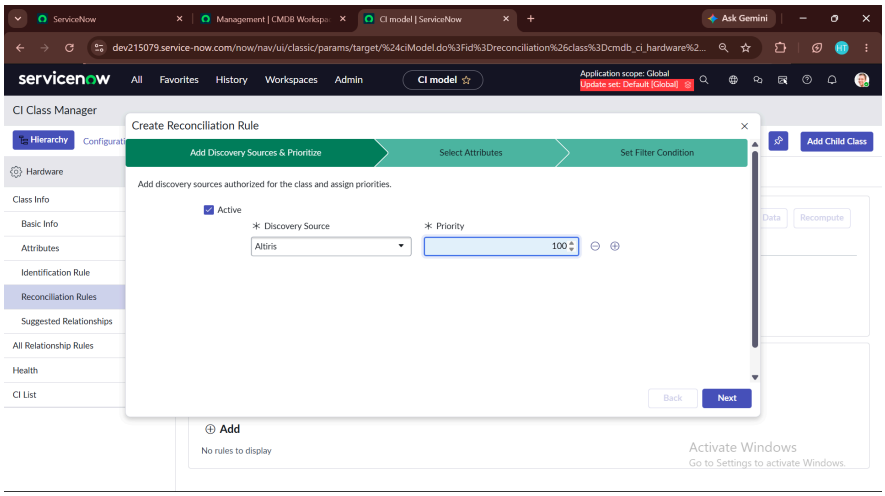
Example

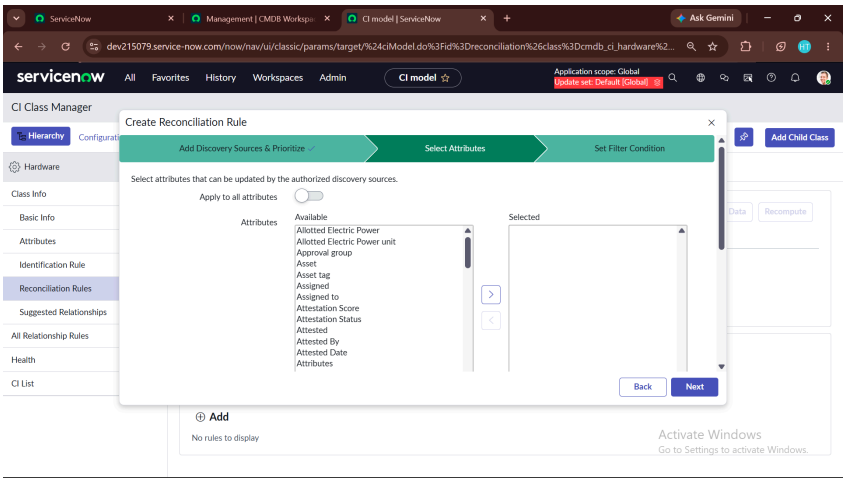
Condition: *Install status is not Retired*.

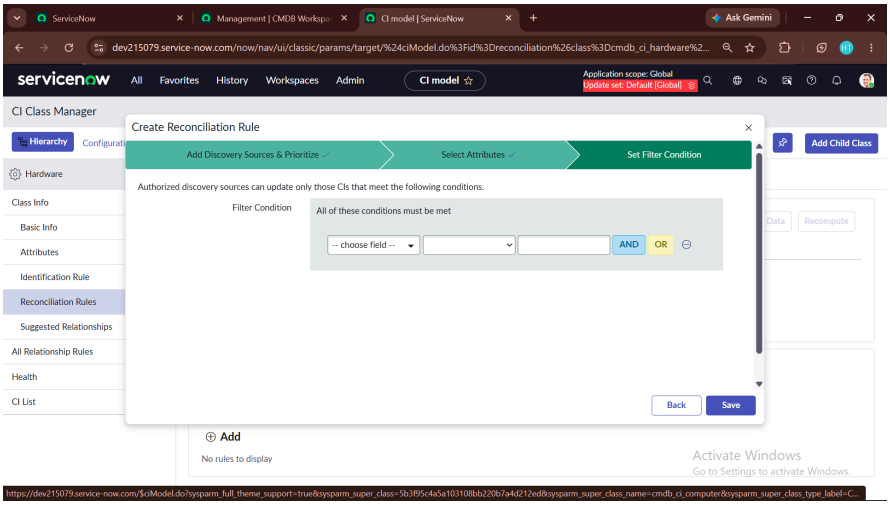
- A new device arrives with serial number **ABC123**.
- An old **Retired** CI has the same serial number.
- IRE **ignores the retired CI**, so it creates a new active CI (or matches an active one) instead of reviving the retired record.

When to use it: to stop IRE from matching against old, retired or test CIs.


ServiceNow
Management | CMDB Workspace
CI model | ServiceNow
Ask Gemini
dev215079.service-now.com/now/nav/ui/classic/params/target/%24ciModel.do%3Fid%3Dreconciliation%26class%3Dcmdb_ci_hardware%2...
servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Global
Update set: Default [Global]
CI Class Manager
Hierarchy
Configuration Item > Hardware
Add Child Class
Hardware
Reconciliation Rules
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Reconciliation Rules
Determine which discovery sources are allowed to populate attributes and in which order.
Preview Rule
Preview Data
Recompute
Created (0)
Add
No rules to display
Data Refresh Rules
Specify when a CI turns stale for a discovery source, allowing the CI to be updated by a lower-priority authorized discovery source.
Created (0)
Add
No rules to display
Activate Windows
Go to Settings to activate Windows.


ServiceNow
Management | CMDB Workspace
CI model | ServiceNow
Ask Gemini
dev215079.service-now.com/now/nav/ui/classic/params/target/%24ciModel.do%3Fid%3Dreconciliation%26class%3Dcmdb_ci_hardware%2...
servicenow
All
Favorites
History
Workspaces
Admin
CI model
Application scope: Global
Update set: Default [Global]
CI Class Manager
Hierarchy
Add Child Class
Hardware
Class Info
Basic Info
Attributes
Identification Rule
Reconciliation Rules
Suggested Relationships
All Relationship Rules
Health
CI List
Create Reconciliation Rule
Add Discovery Sources & Prioritize
Select Attributes
Set Filter Condition
Add discovery sources authorized for the class and assign priorities.
Active
* Discovery Source
Altiris
* Priority
100
Back
Next
Add
No rules to display
Activate Windows
Go to Settings to activate Windows.

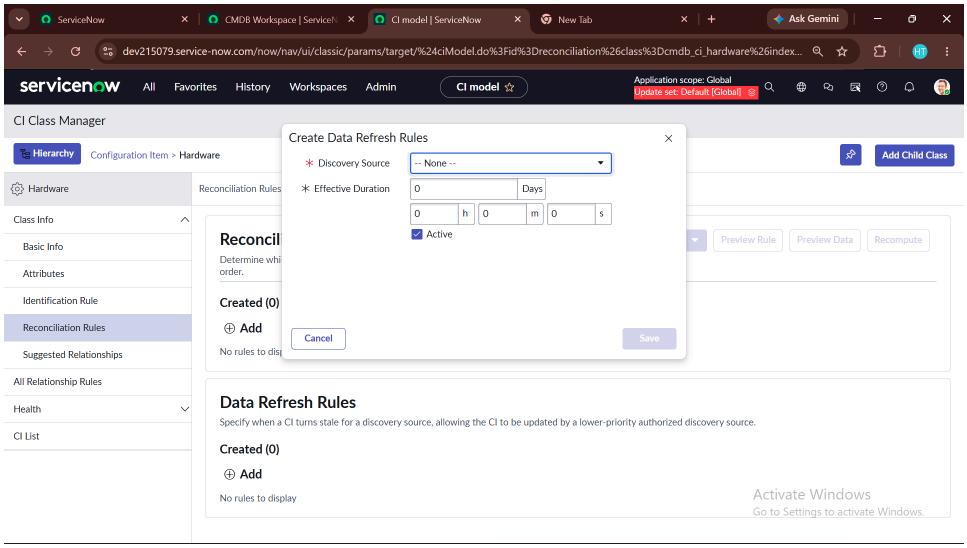




Reconciliation Rules decide **which data source wins** when two sources send different values for the same CI field, and in what order.

Example: Discovery says Laptop-01's *Location* is "Delhi". A manual import says "Mumbai".

- Rule: *Discovery* has higher priority than *Import* for the *Location* field.
- Result: Location stays **Delhi**, and the import cannot overwrite it.



Data Refresh Rules define when a source's data for a CI is considered stale, so a lower-priority source is then allowed to update it.

Example:

- Discovery is the top-priority source for Laptop-01.
- Rule: *Discovery data turns stale after 30 days*.
- Discovery stops updating Laptop-01, so after 30 days the Import source can update it.

Without this rule, a high-priority source blocks lower ones forever, even if it has stopped sending data.

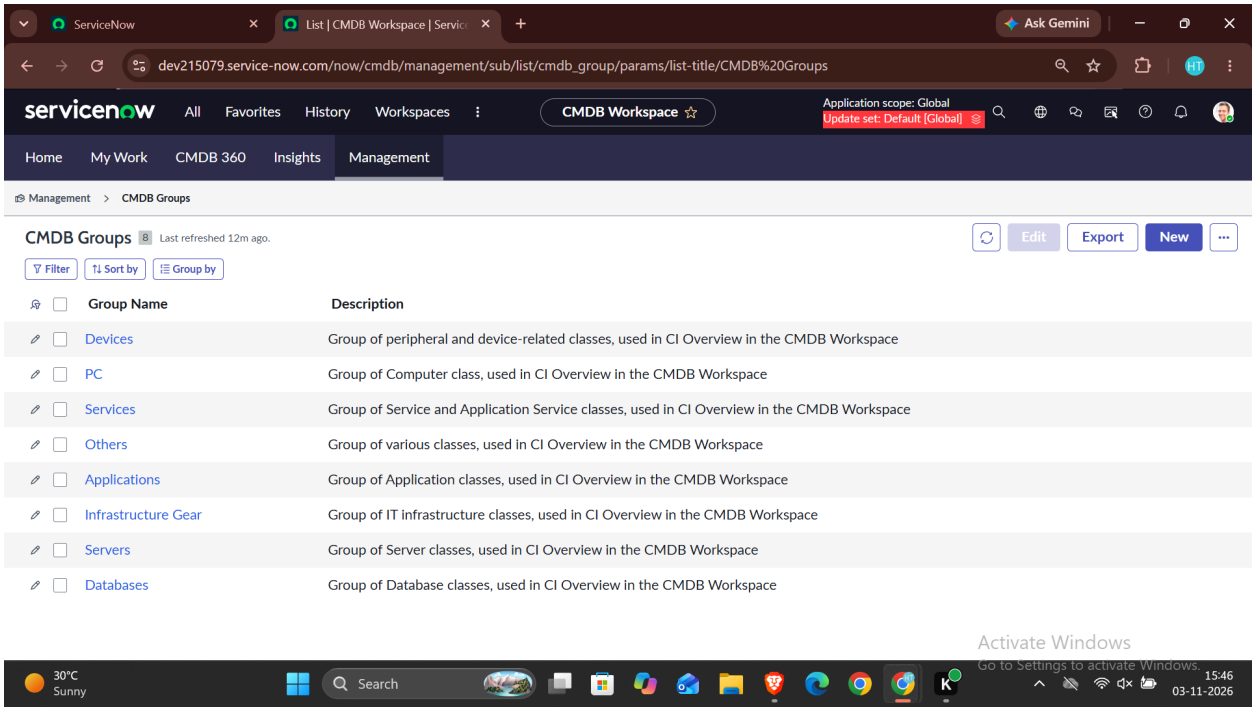
2. **CMDB Groups :**

CMDB Groups are named groups of CI classes. The workspace uses them to **organize CIs in the CI overview chart** on Home.

Example: the **PC** group holds the Computer class, and the **Servers** group holds the Server classes. That's why the Home chart shows bars named PC, Servers, Services, Devices, and so on.

Your list (8 groups): Devices, PC, Services, Others, Applications, Infrastructure Gear, Servers, Databases.

Use: open a group (or click **New**) to add or remove classes, so you control how CIs are grouped and counted.



CSDM

CSDM (Common Service Data Model) is ServiceNow's standard blueprint for organizing CMDB data, so services, applications and infrastructure are linked in a consistent way.

Example chain: Business Service (Online Banking) → Application Service → Application (SAP) → Server (SAP-WEB01).

Use: it makes impact analysis clear. If SAP-WEB01 fails, you can see which service and business is affected.

You saw it earlier as CSDM Data Foundations Dashboard in Management.

CSDM is adopted in phases:

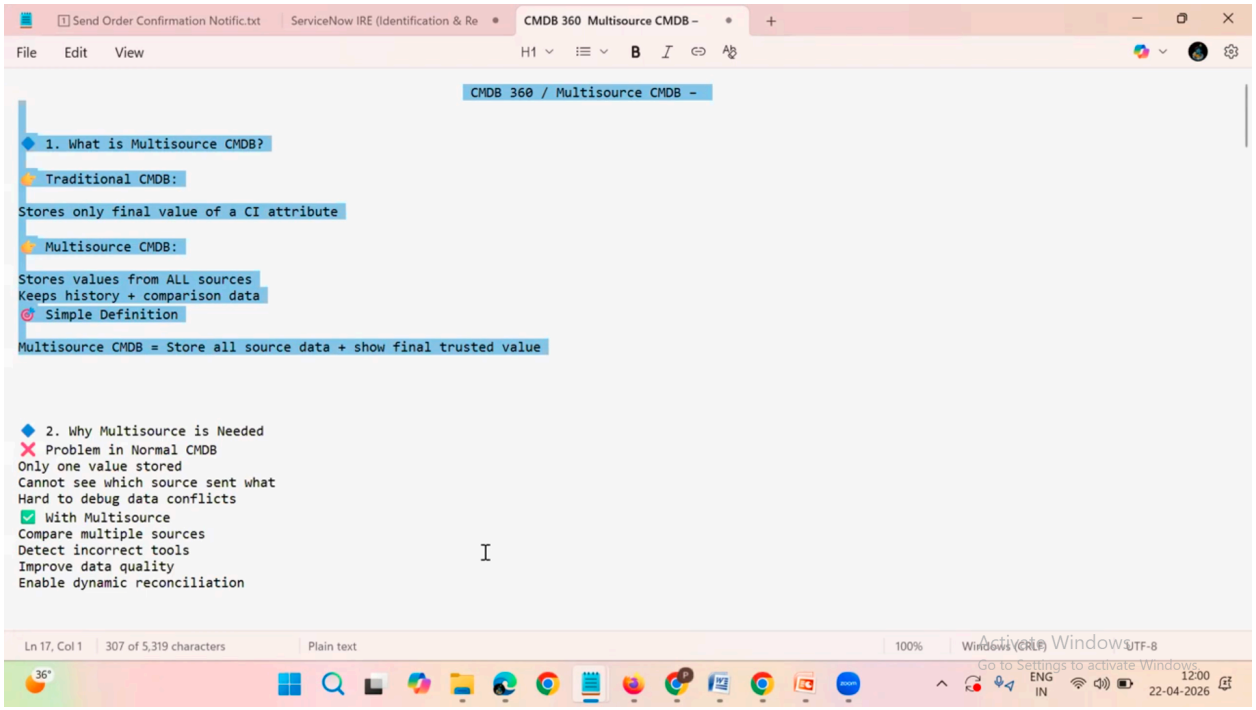
Phase	What you set up
Foundation	Base data: Companies, Locations, Users, Groups, Departments
Crawl	Core CIs: Business Applications, Application Services, Servers, with basic ownership
Walk	Link them: Technical Services, Service Offerings, dependencies between services and CIs
Run	Advanced use: business capabilities, service portfolio, full service views for planning and reporting

(Newer CSDM versions may add more detailed domains, but these are the core phases.)

Simple idea: start with the basics (Foundation), then add CIs (Crawl), connect them (Walk), and use them for business decisions (Run).

The CSDM Data Foundations Dashboard in Management shows how far you have progressed through these phases.

CMDB 360



Send Order Confirmation Notific.txt | ServiceNow IRE (Identification & Re | CMDB 360 Multisource CMDB –

File Edit View H1

CMDB 360 / Multisource CMDB –

🔷 1. What is Multisource CMDB?

👉 Traditional CMDB:

Stores only final value of a CI attribute

👉 Multisource CMDB:

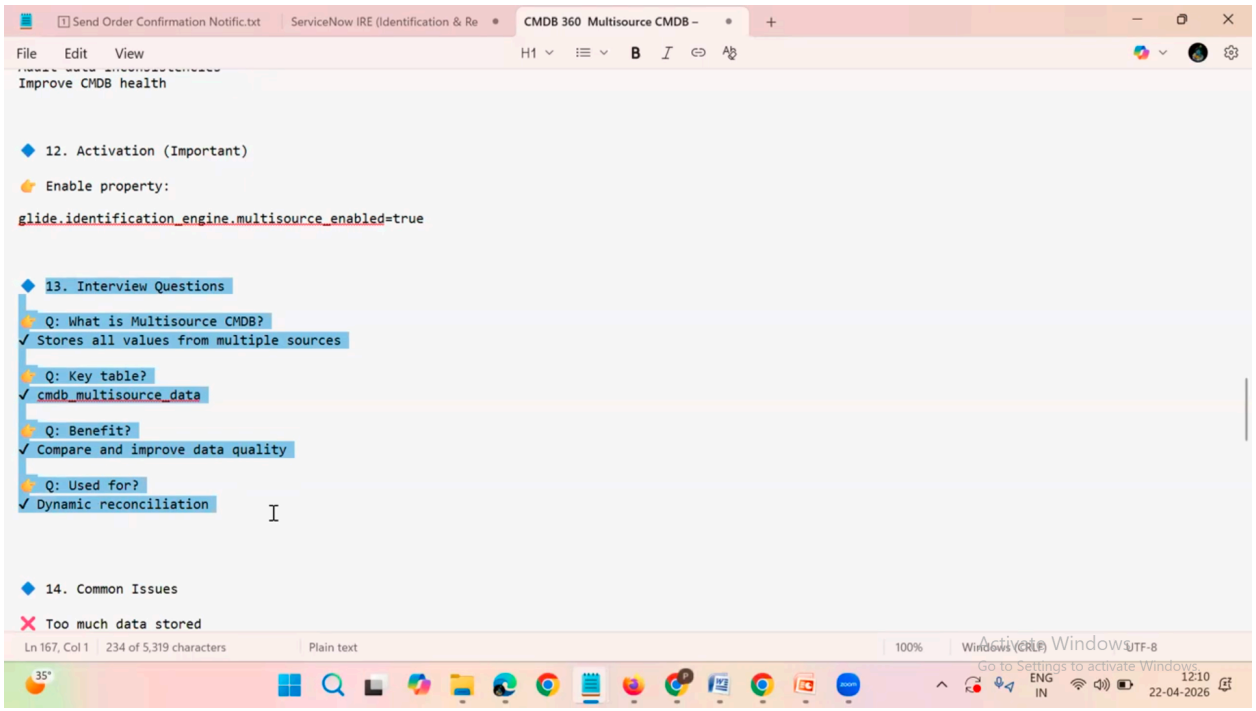
Stores values from ALL sources
Keeps history + comparison data
🎯 Simple Definition

Multisource CMDB = Store all source data + show final trusted value

🔷 2. Why Multisource is Needed
❌ Problem in Normal CMDB
Only one value stored
Cannot see which source sent what
Hard to debug data conflicts
✅ With Multisource
Compare multiple sources
Detect incorrect tools
Improve data quality
Enable dynamic reconciliation

Ln 17, Col 1 | 307 of 5,319 characters | Plain text | 100% | Windows (CRLF) | UTF-8

Activate Windows
Go to Settings to activate Windows.



Send Order Confirmation Notific.txt | ServiceNow IRE (Identification & Re | CMDB 360 Multisource CMDB –

File Edit View H1

Improve CMDB health

🔷 12. Activation (Important)

👉 Enable property:

glide.identification_engine.multisource_enabled=true

🔷 13. Interview Questions

👉 Q: What is Multisource CMDB?
✓ Stores all values from multiple sources

👉 Q: Key table?
✓ cmdb_multisource_data

👉 Q: Benefit?
✓ Compare and improve data quality

👉 Q: Used for?
✓ Dynamic reconciliation

🔷 14. Common Issues

❌ Too much data stored

Ln 167, Col 1 | 234 of 5,319 characters | Plain text | 100% | Windows (CRLF) | UTF-8

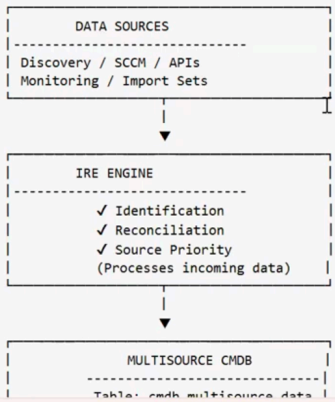
Activate Windows
Go to Settings to activate Windows.

Send Order Confirmation Notific.txt | ServiceNow IRE (Identification & Re | CMDB 360 Multisource CMDB –

File Edit View H1

🔥 Golden Line

"Multisource CMDB provides full visibility of all data sources before deciding the final trusted value."

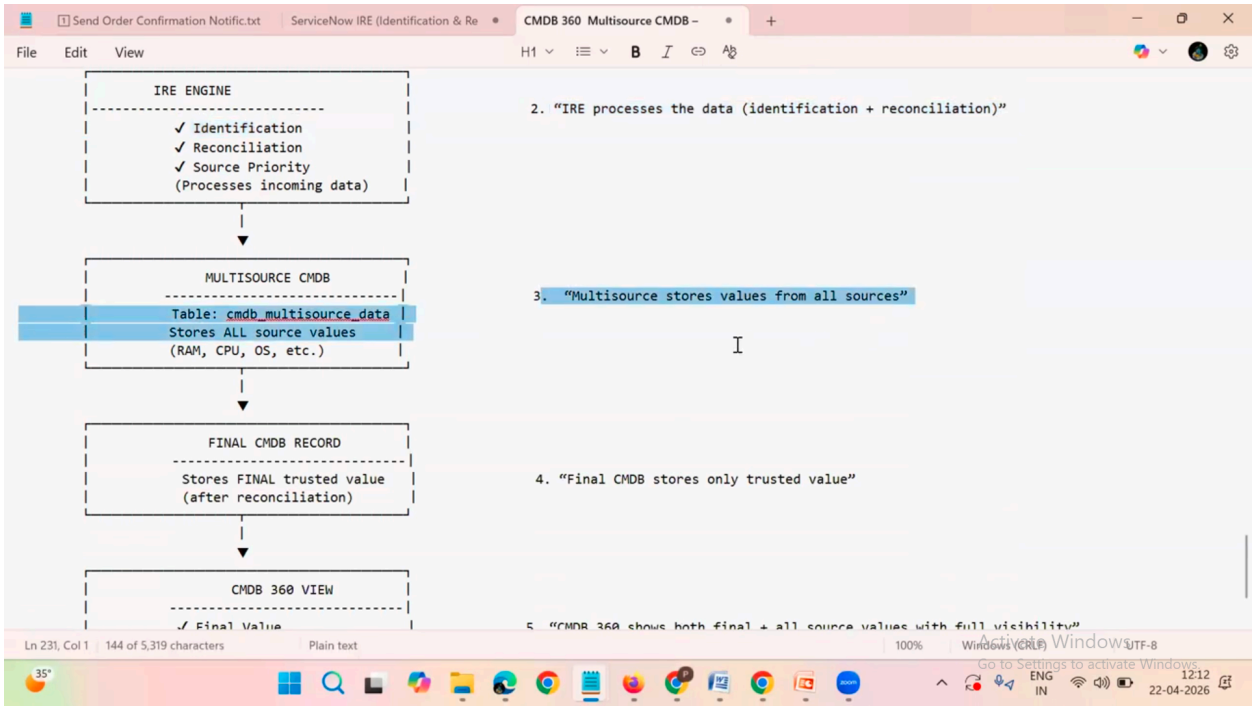


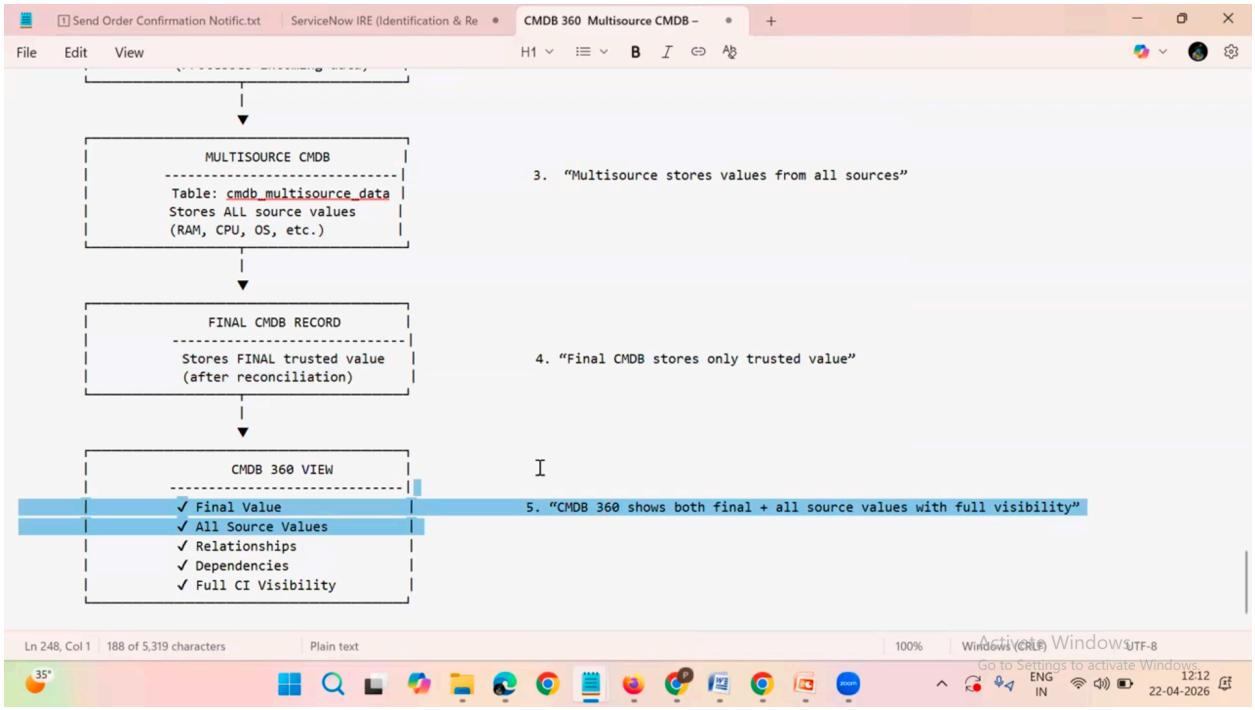
1. "Multiple tools send CI data"
2. "IRE processes the data (identification + reconciliation)"
3. "Multisource stores values from all sources"

Ln 204, Col 105 | 107 of 5,319 characters | Plain text | 100% | Windows (CRLF) | UTF-8

Activate Windows
Go to Settings to activate Windows.

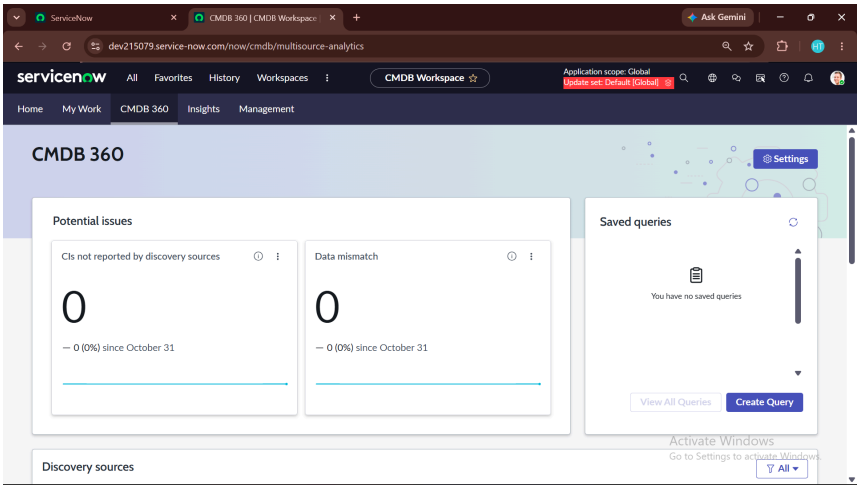
35° ENG IN 12:12 22-04-2026

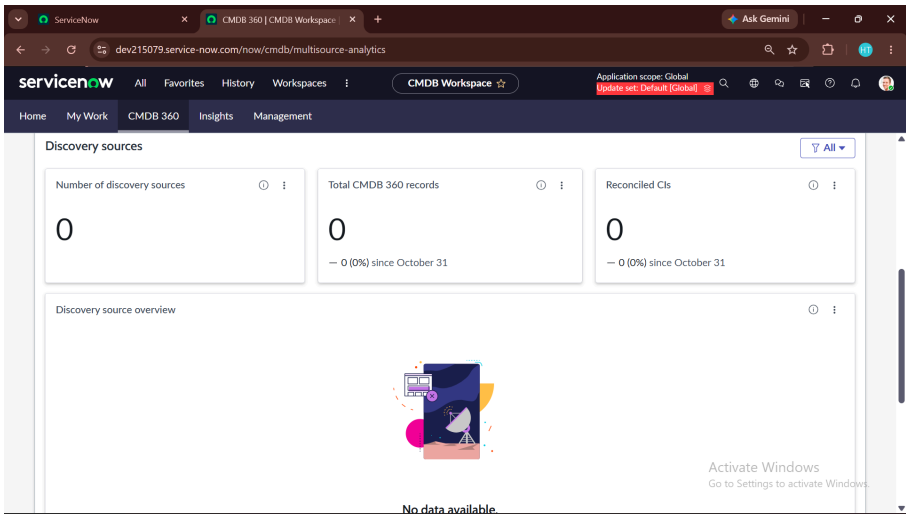

Send Order Confirmation Notific.txt
ServiceNow IRE (Identification & Re
CMDB 360 Multisource CMDB –
File
Edit
View
IRE ENGINE
✓ Identification
✓ Reconciliation
✓ Source Priority
(Processes incoming data)
MULTISOURCE CMDB
Table: cmdb_multisource_data
Stores ALL source values
(RAM, CPU, OS, etc.)
FINAL CMDB RECORD
Stores FINAL trusted value
(after reconciliation)
CMDB 360 VIEW
2. "IRE processes the data (identification + reconciliation)"
3. "Multisource stores values from all sources"
4. "Final CMDB stores only trusted value"
Ln 231, Col 1
144 of 5,319 characters
Plain text
100%
Windows (CRLF)
UTF-8
Activate Windows
Go to Settings to activate Windows.
ENG
IN
12:12
22-04-2026

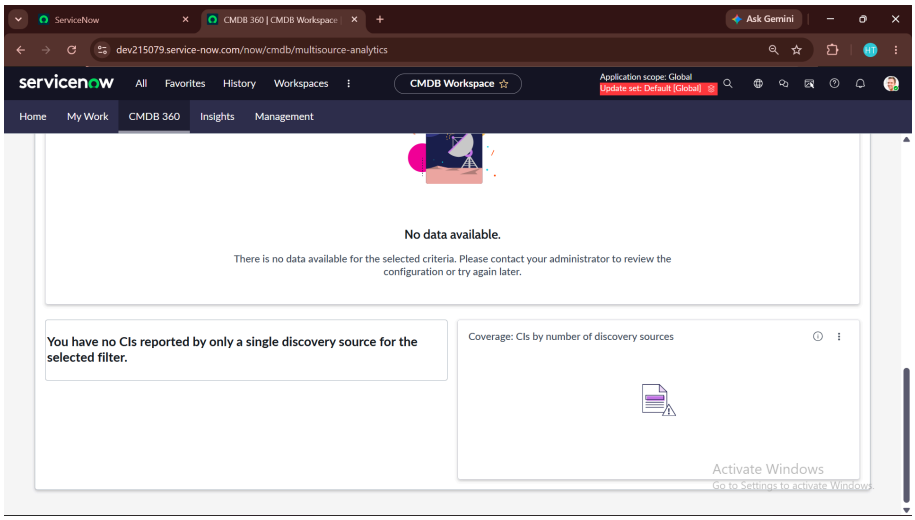

Send Order Confirmation Notific.txt
ServiceNow IRE (Identification & Re
CMDB 360 Multisource CMDB –
File
Edit
View
MULTISOURCE CMDB
Table: cmdb_multisource_data
Stores ALL source values
(RAM, CPU, OS, etc.)
FINAL CMDB RECORD
Stores FINAL trusted value
(after reconciliation)
CMDB 360 VIEW
✓ Final Value
✓ All Source Values
✓ Relationships
✓ Dependencies
✓ Full CI Visibility
3. "Multisource stores values from all sources"
4. "Final CMDB stores only trusted value"
5. "CMDB 360 shows both final + all source values with full visibility"
Ln 248, Col 1
188 of 5,319 characters
Plain text
100%
Windows (CRLF)
UTF-8
Activate Windows
Go to Settings to activate Windows.
ENG
IN
12:12
22-04-2026

To enable cmdb 360, follow two steps :

1. Install ITOM Discovery License Plugin
2. Set System Property glide.identification_engine.multisource_enabled as true







Servicenow **CMDB 360** (formerly called *Multisource CMDB*) tracks data coming into your Configuration Management Database (CMDB) from **multiple tools** (like AWS, Jamf, Microsoft Intune, ServiceNow Discovery, SCCM) so you can compare and clean up duplicate or conflicting information.

The Real-World Analogy: **Your Contact List**

Imagine creating a single contact entry for a friend named **Alex**, but getting information from 3 sources:

* **WhatsApp** says Alex’s phone is `555-1111` and email is `alex@gmail.com`.

* **LinkedIn** says Alex’s job title is `Senior Engineer` and email is `alex@work.com`.

* **Google Contacts** says Alex’s phone is `555-2222`.

Here is what each component shown in your CMDB 360 dashboard means:

1. **Configuration Item (CI)**

* **What it means:** An asset or component tracked in your IT system (e.g., a laptop, server, application, or database).

* **Simple Example:** **"Alex"** is the CI.

2. **Discovery Sources**

* **What it means:** The background tools feeding data into ServiceNow.

* **Simple Example:** **WhatsApp**, **LinkedIn**, and **Google Contacts** are your discovery sources.

3. **CIs Not Reported by Discovery Sources**

* **What it means:** CIs registered in ServiceNow that haven't been updated or seen by any discovery tool during a given timeframe.

* **Simple Example:** A friend named **"John"** is in your address book, but neither WhatsApp, LinkedIn, nor Google Contacts has provided any updates or activity for him lately.

In short: An asset exists in your ServiceNow list, but no discovery tools (like SCCM or AWS) have reported or seen it active recently.

Example: John is in your contact list, but none of your apps have seen him online in months.

It means the record is **stale** or **ghost data**—it exists on paper in your CMDB, but no live system has confirmed it's actually running or connected.

4. **Data Mismatch**

* **What it means:** Different discovery tools report conflicting details about the exact same asset.

* **Simple Example:** WhatsApp says Alex's phone number is `555-1111`, but Google Contacts says it is `555-2222`. CMDB 360 flags this as a **Data Mismatch** so you can fix it.

5. **Total CMDB 360 Records**

* **What it means:** The total count of data records collected across all discovery sources for all CIs.

* **Simple Example:** The total count of raw details collected (3 source entries for Alex + 1 entry for John = 4 records).

6. **Reconciled CIs**

* **What it means:** Assets where ServiceNow successfully resolved conflicting attributes using priority rules (reconciliation rules) to pick the single "trusted" truth.

* **Simple Example:** You set a rule stating *"Always trust LinkedIn for emails, but trust WhatsApp for phone numbers."* ServiceNow automatically picks `alex@work.com` and `555-1111`. Alex is now a **Reconciled CI**.

7. **Coverage: CIs by Number of Discovery Sources**

* **What it means:** A metric showing how many tools report on each asset (e.g., how many CIs are reported by 1 source vs. 2 sources vs. 3 sources).

* **Simple Example:**

* **Alex** is covered by **3 sources** (High visibility).

* **John** is covered by **1 source** (Low visibility/risk of outdated data)

8. Discovery source overview: It shows a visual Breakdown of **which specific tools** (like AWS, Jamf, SCCM, or ServiceNow Discovery) are bringing data into your CMDB, and **how much** data each tool is contributing.

Example: A pie chart showing:

- **ServiceNow Discovery:** 50%
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- **AWS Import:** 30%
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- **SCCM:** 20%